10 verified against NCERT

14 concept & approach

PHYSICAL 4

2015 A Q5 ●

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BIOLOGY 10

2015 A Q59 ● 2015 A Q63 ● 2015 A Q90 ● 2015 A Q107 ● 2015 A Q116 ● 2015 A Q120 ● 2015 A Q127 ● 2016 X Q55 ● 2016 X Q100 ● 2016 X Q126 ●

PHYSICS 10

2015 A Q136 ●

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2015 A Q143 ●

2015 A Q144 ●

2015 A Q146 ●

2015 A Q147 ●

2015 Code A **Q5** CHEMISTRY / PHYSICAL

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physical	Chemical Equilibrium	Law of Chemical Equilibrium	easy	3

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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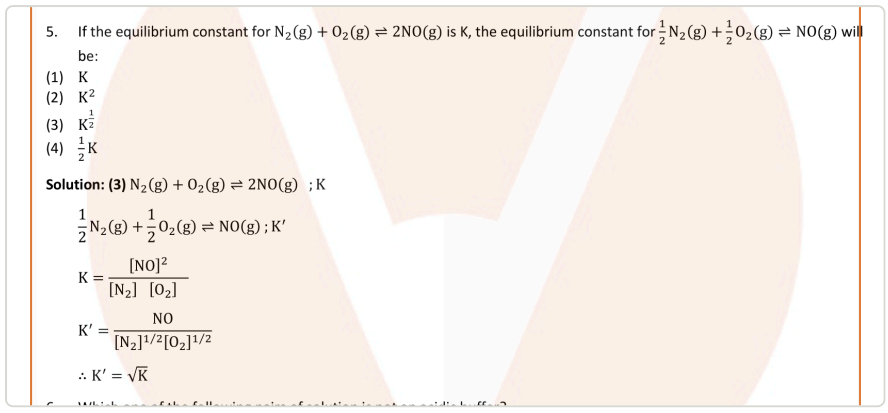

5. If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be:

(1) K
(2) K^2
(3) $K^{\frac{1}{2}}$
(4) $\frac{1}{2}K$

Solution: (3) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$; K

$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$; K′

$K = \frac{[NO]^2}{[N_2]\ [O_2]}$

$K' = \frac{NO}{[N_2]^{1/2}[O_2]^{1/2}}$

$\therefore K' = \sqrt{K}$

READ FROM THE IMAGE BY B3-QUESTION-READER

If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be:

(1) K

(2) K^2

(3) $K^{\frac{1}{2}}$ ←printed answer

(4) $\frac{1}{2}K$

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT

K' = K^n when every coefficient of a reaction is multiplied by n

APPROACH

Write the Kc expression for each equation; the second is the first with all coefficients halved, so the scaling rule for equilibrium constants gives the new constant from K.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Chemistry
STREAM	physical
CHAPTER	Chemical Equilibrium HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on how the equilibrium constant transforms when a balanced equation is divided by two, i.e. K' = K^n.
SUBTOPIC 1	Law of Chemical Equilibrium
WHY THIS SUBTOPIC	The answer depends on the equilibrium constant being defined by the equation as written, so halving the stoichiometric coefficients gives K^1/2.
DIFFICULTY	**easy** · 1 step
STEPS	recall that scaling a balanced equation by a factor n raises the equilibrium constant to the power n, and apply it with n = 1/2
WHY	The whole task is one application of the K' = K^n rule to a halved equation; no concentrations or stoichiometric arithmetic are involved.
THE CATCH	Options 2 and 4 are the exact results of the two standard wrong routes - squaring instead of taking the root, and dividing K by two as if K scaled linearly with the coefficients.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`derive_expression`

MEANS a symbolic expression

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Scaling a balanced equation scales its equilibrium constant as a power: halve the coefficients and*
>
> K
>
> *becomes*
>
> $K^{1/2}$

E4 Thinking

three separate questions, not one

`application`

MEANS a stated rule, criterion or convention applied to this case. The rule must be known first — application is not knowledge-free

`single_step`

MEANS one step, however much knowledge it needs

E5 Trap

one option, one mistake, or nothing

OPTION 1 **`wrong_power_of_dependence`**

MEANS linear where it should be squared, squared where it should be $\sqrt{ed}$, or a root dropped

WHY HERE K is taught as a constant of the reaction at a fixed temperature, so a student who takes "the same reaction" at its word leaves the constant untouched. Halving every coefficient takes each concentration term to its square root, so the exponent is 1/2 and not 1.

OPTION 2 **`wrong_power_of_dependence`**

MEANS linear where it should be squared, squared where it should be $\sqrt{ed}$, or a root dropped

WHY HERE Doubling an equation squares its constant is the worked case in the chapter, so a student who keeps the pairing "factor of 2 goes with power 2" but not which way the printed change runs reaches for K^2. The printed change is a halving, so the exponent is 1/2; K^2 is what the reverse scaling gives.

OPTION 4 **`calculation_slip`**

MEANS a specific wrong operation yields a printed option

WHY HERE Carrying the $\frac{1}{2}$ from the coefficients straight onto K as a multiplier gives $\frac{1}{2}K$, the result of scaling the constant linearly with the equation. The coefficients enter K as exponents on the concentrations, so a halved coefficient is a halved power.

2015 Code A **Q6** CHEMISTRY / PHYSICAL

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physical	Ionic Equilibrium	Buffer Solutions	medium	3

1 The question

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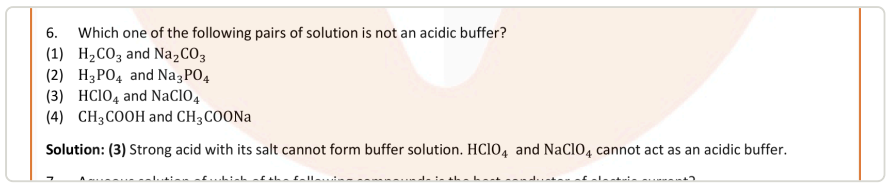

6. Which one of the following pairs of solution is not an acidic buffer?
(1) H_2CO_3 and Na_2CO_3
(2) H_3PO_4 and Na_3PO_4
(3) $HClO_4$ and $NaClO_4$
(4) CH_3COOH and CH_3COONa

Solution: (3) Strong acid with its salt cannot form buffer solution. $HClO_4$ and $NaClO_4$ cannot act as an acidic buffer.

READ FROM THE IMAGE BY B3-QUESTION-READER

Which one of the following pairs of solution is not an acidic buffer?

(1) H_2CO_3 and Na_2CO_3

(2) H_3PO_4 and Na_3PO_4

(3) $HClO_4$ and $NaClO_4$ ←printed answer

(4) CH_3COOH and CH_3COONa

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **an acidic buffer is a weak acid plus the salt of its conjugate base**

APPROACH Classify the acid in each pair as weak or strong; the pair built on a strong acid has no reservoir of undissociated acid and so cannot buffer.

ALSO USES **pH = pKa + log([salt]/[acid])**

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Chemistry
STREAM	physical
CHAPTER	Ionic Equilibrium HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the definition of an acidic buffer as a weak acid with its conjugate salt, so the strong acid HClO4 with NaClO4 fails - a buffer-solution fact from ionic equilibrium.
SUBTOPIC 1	Buffer Solutions
WHY THIS SUBTOPIC	The answer depends on the composition rule for an acidic buffer - a weak acid plus its conjugate salt - which the strong acid HClO4 with NaClO4 violates.
DIFFICULTY	**medium** · 2 steps
STEPS	recall that an acidic buffer is a weak acid together with the salt of that weak acid → classify the acid in each pair as weak or strong and find the pair built on a strong acid (HClO4)
WHY	The work is one definition recalled and then applied as a strong/weak classification over the four acid-salt pairs.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 **Question type**

the shape on the page

`pair_options`

MEANS each option is a pair, group or set of same-kind items accepted or rejected whole — half an option right scores zero

E2 **Format**

what the ANSWER must contain

`pick_both_parts`

MEANS two or more independent answers that must ALL hold at once: magnitude and direction, two numbers, a pair or triple of names. Half right scores nothing

E3 **Content shape**

what is being tested, plus the actual concept

`definition_term`

MEANS what a term or quantity means, including holding two neighbouring terms apart; the strict reading of the definition decides it

> *An acidic buffer is a WEAK acid together with its own salt; a strong acid and its salt cannot buffer*

E4 **Thinking**

three separate questions, not one

`application`

MEANS a stated rule, criterion or convention applied to this case. The rule must be known first — application is not knowledge-free

`several_independent_checks`

MEANS each item verified separately, verdicts combined

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **`true_statement_trap`**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE The carbonic acid buffer is drilled as H_2CO_3 with $NaHCO_3$, so a student who checks the printed salt against that memory finds Na_2CO_3 instead and rules the pair out. H_2CO_3 is still a weak acid and Na_2CO_3 still one of its salts, so the pair does buffer and the negated task is not answered by it.

OPTION 2 **`true_statement_trap`**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE A solution of Na_3PO_4 is strongly alkaline, so a student who judges the mixture by the salt's own reaction calls it basic rather than an acidic buffer. The pair is still a weak acid with its conjugate base, which is what an acidic buffer is made of; what disqualifies option 3 is that $HClO_4$ ionises completely.

2015 Code A **Q7** CHEMISTRY / PHYSICAL

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physical	Electrochemistry	Weak Electrolytes	easy	4

1 The question

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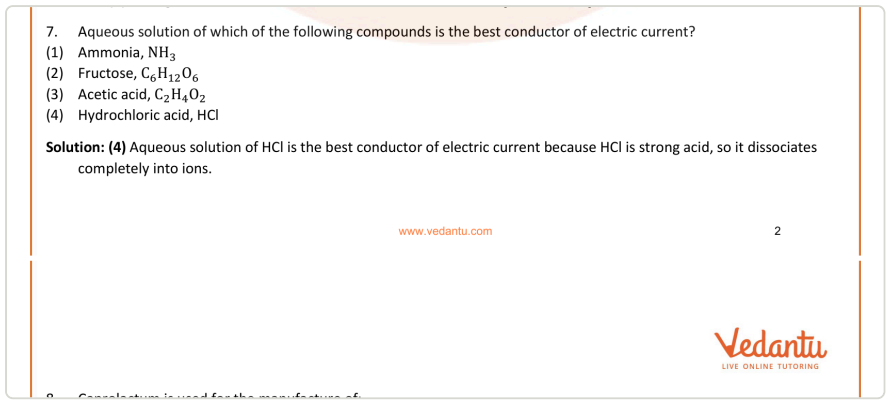

7. Aqueous solution of which of the following compounds is the best conductor of electric current?
(1) Ammonia, NH_3
(2) Fructose, $C_6H_{12}O_6$
(3) Acetic acid, $C_2H_4O_2$
(4) Hydrochloric acid, HCl

Solution: (4) Aqueous solution of HCl is the best conductor of electric current because HCl is strong acid, so it dissociates completely into ions.

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READ FROM THE IMAGE BY B3-QUESTION-READER

Aqueous solution of which of the following compounds is the best conductor of electric current?

(1) Ammonia, NH_3

(2) Fructose, $C_6H_{12}O_6$

(3) Acetic acid, $C_2H_4O_2$

(4) Hydrochloric acid, HCl ←printed answer

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT

conductivity rises with the concentration of free ions in solution

APPROACH

Sort the solutes into non-electrolyte, weak electrolyte and strong electrolyte; the degree of dissociation decides how many ions each releases and so which solution conducts best.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
4	4 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Chemistry
STREAM	physical
CHAPTER	Electrochemistry HIGH CONFIDENCE
WHY THIS CHAPTER	The answer turns on HCl being a strong electrolyte that ionises completely and therefore conducts best, which is the strong-versus-weak-electrolyte and conductance of electrolytic solutions material of Electrochemistry.
SUBTOPIC 1	Weak Electrolytes
SUBTOPIC 2	Some Important Terms Related to Conductance
WHY THIS SUBTOPIC	The answer depends on classifying the solutes as strong electrolyte (HCl, fully ionised, highest ion concentration), weak electrolytes (NH3, acetic acid) and non-electrolyte (fructose), and relating that to conductivity.
DIFFICULTY	**easy** · 1 step
STEPS	recall that conductivity tracks the degree of ionisation, so the only strong electrolyte in the list (HCl) conducts best
WHY	One recall of the strong/weak/non-electrolyte classification applied to four named compounds.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 **Question type**

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 **Format**

what the ANSWER must contain

`name_or_identify`

MEANS a name, term, colour, organism, scientist

E3 **Content shape**

what is being tested, plus the actual concept

`trend`

MEANS behaviour across a series, or a comparison between named cases: which is larger, which way a quantity moves, where the series breaks

> *Strong, weak and non-electrolytes: how many ions a solution carries is what decides its conductivity*

E4 **Thinking**

three separate questions, not one

`application`

MEANS a stated rule, criterion or convention applied to this case. The rule must be known first — application is not knowledge-free

`several_independent_checks`

MEANS each item verified separately, verdicts combined

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **`true_but_out_of_scope`**

MEANS the option is correct subject matter that answers a different question than the stem asked; it fails on relevance, not on truth

WHY HERE Ammonia is the one solute here a student is taught to write as ionising in water without being an acid, $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$, so "which of these makes ions" lands on it. It is a weak base and only a small fraction of it ionises, so its solution carries far fewer ions than one of hydrochloric acid.

OPTION 2 **`no_derivation_route`**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE Fructose dissolves freely but as whole molecules, and the chapter's one rule for conduction - a solution conducts through its ions - gives it none, so no reading of that rule reaches this option. It is the set's non-electrolyte, which is what makes the other three a graded series.

OPTION 3 **`true_but_out_of_scope`**

MEANS the option is correct subject matter that answers a different question than the stem asked; it fails on relevance, not on truth

WHY HERE Acetic acid is the acid used to illustrate conduction by a weak electrolyte, so a student who sorts the four by "is it an acid" and takes the one the conductivity discussion names stops here. It answers "which of these conducts" rather than the stem's "best": about one molecule in a hundred ionises, against complete ionisation for HCl.

2015 Code A **Q12** CHEMISTRY / PHYSICAL

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physical	Basic Concepts of Chemistry	Mole Concept	easy	2

1 The question

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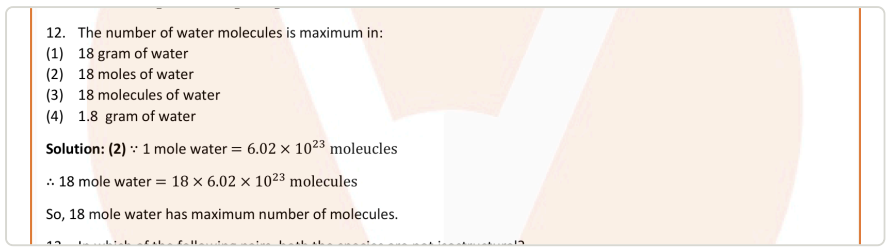

12. The number of water molecules is maximum in:
(1) 18 gram of water
(2) 18 moles of water
(3) 18 molecules of water
(4) 1.8 gram of water

Solution: (2) ∵ 1 mole water = 6.02×10^{23} moleucles

∴ 18 mole water = $18 \times 6.02 \times 10^{23}$ molecules

So, 18 mole water has maximum number of molecules.

READ FROM THE IMAGE BY B3-QUESTION-READER

The number of water molecules is maximum in:

(1) 18 gram of water

(2) 18 moles of water ←printed answer

(3) 18 molecules of water

(4) 1.8 gram of water

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **N = n N_A, with n = m / M**

APPROACH Put every option on the same footing in moles - mass divided by molar mass, moles as stated, molecules divided by N_A - then compare the molecule counts.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
2	2 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Chemistry
STREAM	physical
CHAPTER	Basic Concepts of Chemistry HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on converting grams, moles and molecule counts to a common basis via the Avogadro number, i.e. the mole concept.
SUBTOPIC 1	Mole Concept
WHY THIS SUBTOPIC	The answer depends on converting grams, moles and a raw molecule count into molecules via the Avogadro number, so 18 mol (18 NA) wins.
DIFFICULTY	**easy** · 1 step
STEPS	normalise each option to a molecule count using the mole relation and pick the largest (18 moles)
WHY	One recall of what a mole is, applied to four quantities whose ordering is then immediate.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 **Question type**

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 **Format**

what the ANSWER must contain

`calculate_value`

MEANS a number the student computes from the stem's numbers

E3 **Content shape**

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Mole concept: converting between grams, moles and molecule counts for water, $M = 18$ g/mol*

E4 **Thinking**

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`several_independent_checks`

MEANS each item verified separately, verdicts combined

`estimation`

`unit_sanity`

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **`wrong_quantity`**

MEANS the option answers for a neighbouring PHYSICAL quantity: diameter for radius, moment of inertia for radius of gyration

WHY HERE 18 is the molar mass of water, so a student who sees 18 g and thinks "that is the mole-sized amount" picks the mass line. It is one mole, 6.0×10^{23} molecules, against 1.08×10^{25} for 18 moles: mass answered where the comparison is in amount.

OPTION 3 **`copied_from_stem`**

MEANS the option repeats a phrase from the stem without answering the question

WHY HERE The stem asks for the number of water molecules and this is the only option that answers in molecules, so matching the stem's own word rather than the sizes lands here. It is 18 molecules against 1.08×10^{25} for 18 moles, the smallest sample on the page.

2015 Code A **Q59** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Principles of Inheritance and Variation	Colour Blindness	hard	2

1 The question

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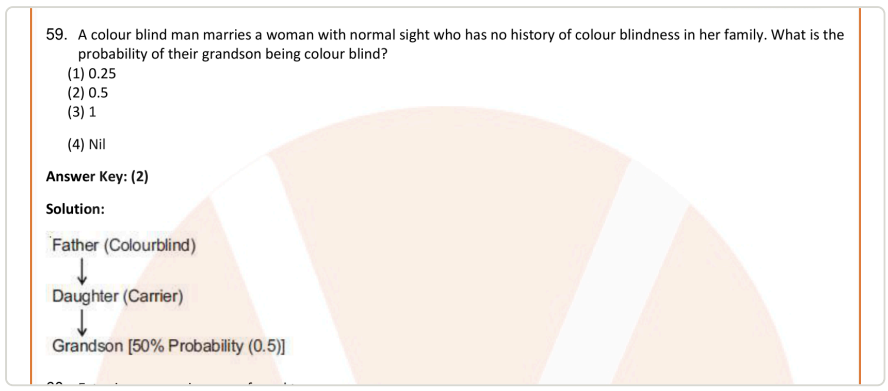

59. A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind?
(1) 0.25
(2) 0.5
(3) 1
(4) Nil

Answer Key: (2)

Solution:

Father (Colourblind)
↓
Daughter (Carrier)
↓
Grandson [50% Probability (0.5)]

READ FROM THE IMAGE BY B3-QUESTION-READER

A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind?

(1) 0.25

(2) 0.5 ←printed answer

(3) 1

(4) Nil

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **X-linked recessive inheritance: an affected father makes every daughter a carrier**

APPROACH Trace the father's X^c to his daughters, who are all carriers, then apply segregation at that carrier's X pair to her sons to get the chance the grandson is affected.

ALSO USES **law of segregation · probabilities along a pedigree path multiply**

how NCERT reaches this answer: **rules chained**

NEW NCERT · BIO12L CH4 · PRINTED PAGE 021

in the chromosome. DNA is the carrier of genetic information. It is hence transmitted from one generation to the other without any change or alteration. However, changes or alteration do take place occasionally. Such an alteration or change in the genetic material is referred to as mutation. A number of disorders in human beings have been found to be associated with the inheritance of changed or altered genes or chromosomes.

4.8.2 Mendelian Disorders

Broadly, genetic disorders may be grouped into two categories – Mendelian disorders and Chromosomal disorders. Mendelian disorders are mainly determined by alteration or mutation in the single gene. These disorders are transmitted to the offspring on the same lines as we have studied in the principle of inheritance. The pattern of inheritance of such Mendelian disorders can be traced in a family by the pedigree analysis. Most common and prevalent Mendelian disorders are Haemophilia, Cystic fibrosis, Sickle-cell anaemia, Colour blindness, Phenylketonuria, Thalassemia, etc. It is important to mention here that such Mendelian disorders may be dominant or recessive. By pedigree analysis one can easily understand whether the trait in question is dominant or recessive. Similarly, the trait may also be linked to the sex chromosome as in case of haemophilia. It is evident that this X-linked recessive trait shows transmission from carrier female to male progeny. A representative pedigree is shown in Figure 4.14 for dominant and recessive traits. Discuss with your teacher and design pedigrees for characters linked to both autosomes and sex chromosome.

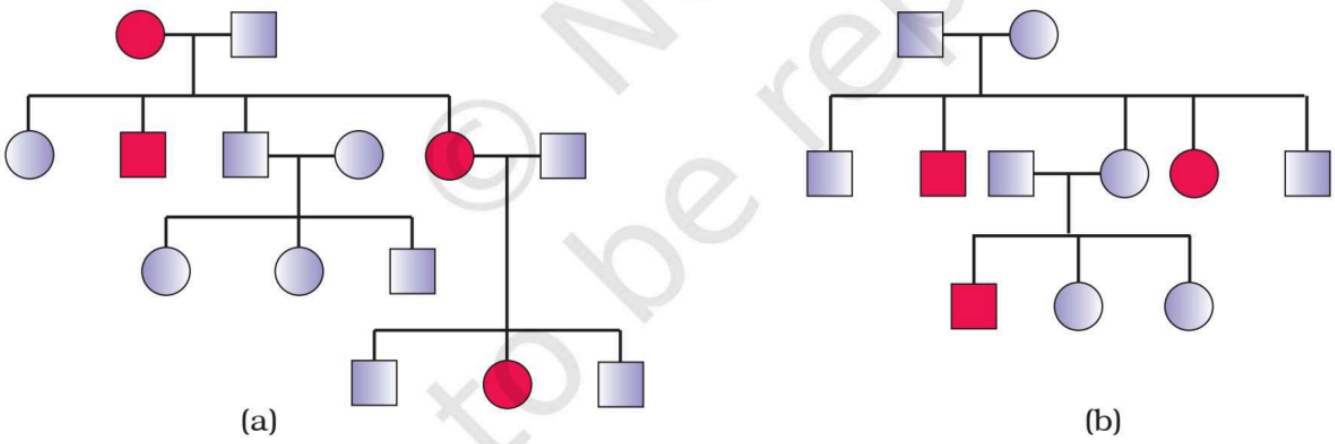

Figure 4.14 Representative pedigree analysis of (a) Autosomal dominant trait (for example: Myotonic dystrophy) (b) Autosomal recessive trait (for example: Sickle-cell anaemia)

Colour Blindness : It is a sex-linked recessive disorder due to defect in either red or green cone of eye resulting in failure to discriminate between red and green colour. This defect is due to mutation in certain genes present in the X chromosome. It occurs in about 8 per cent of males and only about 0.4 per cent of females. This is because the genes that lead to red-green colour blindness are on the X chromosome. Males have only one X chromosome and females have two. The son of a woman who carries

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BIO12L-CH04-P87 · 4.8.2 Mendelian Disorders

> Colour Blindness : It is a sex-linked recessive disorder due to defect in either red or green cone of eye resulting in failure to discriminate between red and green colour. This defect is due to mutation in certain genes present in the X chromosome. It occurs in about 8 per cent of males and only about 0.4 per cent of females. This is because the genes that lead to red-green colour blindness are on the X chromosome. Males have only one X

chromosome and females have two. The son of a woman who carries the gene has a 50 per cent chance of being colour blind. The mother is not herself colour blind because the gene is recessive. That means that its effect is suppressed by her matching dominant normal gene. A daughter will not normally be colour blind, unless her mother is a carrier and her father is colour blind.

HOW THE QUOTED LINES REACH THIS ANSWER

1. The colour blind man is X^c Y; the woman with normal sight and no family history of colour blindness is homozygous normal X^C X^C.
2. The man has only one X and it carries the colour blindness allele, so every F_1 daughter receives that X^c from her father and an X^C from her mother, making every daughter a carrier X^C X^c; every F_1 son receives his father's Y and his mother's X^C, so is X^C Y.
3. A grandson is the son of one of those carrier daughters, and the son of a carrier woman has a 50 per cent chance of being colour blind.
4. Probability of the grandson being colour blind = 50 per cent = 0.5.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
2	2 LOW	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Principles of Inheritance and Variation HIGH CONFIDENCE
WHY THIS CHAPTER	The answer requires tracing an X-linked recessive allele from an affected grandfather through his carrier daughters, i.e. sex-linked inheritance and pedigree probability.
SUBTOPIC 1	Colour Blindness
WHY THIS SUBTOPIC	The answer requires tracing the X-linked recessive colour-blindness allele from the affected grandfather through his obligate-carrier daughters to the grandsons.
DIFFICULTY	**hard** · 4 steps
STEPS	write the parents' genotypes: colour-blind father XcY and a homozygous normal mother XX, since her family has no history → work out the F1: every daughter is a carrier XcX and every son is XY normal → decide which F1 child the grandson can come through - only a carrier daughter can transmit the allele - and cross her with a normal male → compute the fraction of that cross's sons who are affected
WHY	The work is two successive crosses plus the choice of which parental line the grandson must descend through.
THE CATCH	The stem does not say whether the grandson comes through a son or a daughter, and the two readings give two different offered values (counting only daughters' sons versus all grandsons), so a trap option is the exact result of the other reading.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`calculate_value`

MEANS a number the student computes from the stem's numbers

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *X-linked recessive inheritance over two generations: a colour blind father makes every daughter a carrier, and a carrier passes the allele to half her sons*

E4 **Thinking**

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **`calculation_slip`**

MEANS a specific wrong operation yields a printed option

WHY HERE A student who treats the sex of the grandchild as still open multiplies the $\frac{1}{2}$ that a carrier daughter passes on the allele by a further $\frac{1}{2}$ for the child being a boy, and gets 0.25. The stem has already fixed the grandchild as a grandson, so that second factor has been spent.

OPTION 3 **`calculation_slip`**

MEANS a specific wrong operation yields a printed option

WHY HERE The colour blind father has one X and gives it to every daughter, and a student who carries that all-or-none transmission into the next generation has the carrier daughter passing the allele to all her sons, which gives 1. She has two X chromosomes and only one carries the allele, so half her sons receive it and the figure is 0.5.

2015 Code A **Q63** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Principles of Inheritance and Variation	Linkage & Recombination	easy	2

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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63. The term "linkage" was coined by

(1) W. Sutton

(2) T.H. Morgan

(3) T. Boveri

(4) G. Mende

Answer key: (2)

Solution: The term "linkage" was coined by T.H. Morgan.

READ FROM THE IMAGE BY B3-QUESTION-READER

The term ”linkage” was coined by

(1) W. Sutton

(2) T.H. Morgan ←printed answer

(3) T. Boveri

(4) G. Mende

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

how NCERT reaches this answer: **states answer**

Sutton and Boveri argued that the pairing and separation of a pair of chromosomes would lead to the segregation of a pair of factors they carried. Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the **chromosomal theory of inheritance**.

Following this synthesis of ideas, experimental verification of the chromosomal theory of inheritance by Thomas Hunt Morgan and his colleagues, led to discovering the basis for the variation that sexual reproduction produced. Morgan worked with the tiny fruit flies, ***Drosophila melanogaster*** (Figure 4.10), which were found very suitable for such studies. They could be grown on simple synthetic medium in the laboratory. They complete their life cycle in about two weeks, and a single mating could produce a large number of progeny flies. Also, there was a clear differentiation of the sexes – the male and female flies are easily distinguishable. Also, it has many types of hereditary variations that can be seen with low power microscopes.

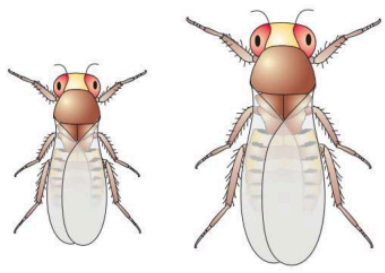

Figure 4.10 *Drosophila melanogaster* (a) Male (b) Female

4.3.3 Linkage and Recombination

Morgan carried out several dihybrid crosses in *Drosophila* to study genes that were sex-linked. The crosses were similar to the dihybrid crosses carried out by Mendel in peas. For example Morgan hybridised yellow-bodied, white-eyed females to brown-bodied, red-eyed males and intercrossed their F_1 progeny. He observed that the two genes did not segregate independently of each other and the F_2 ratio deviated very significantly from the 9:3:3:1 ratio (expected when the two genes are independent).

Morgan and his group knew that the genes were located on the X chromosome (Section 4.4) and saw quickly that when the two genes in a dihybrid cross were situated on the same chromosome, the proportion of parental gene combinations were much higher than the non-parental type. Morgan attributed this due to the physical association or linkage of the two genes and coined the term **linkage** to describe this physical association of genes on a chromosome and the term **recombination** to describe the generation of non-parental gene combinations (Figure 4.11). Morgan and his group also found that even when genes were grouped on the same chromosome, some genes were very tightly linked (showed very low recombination) (Figure 4.11, Cross A) while others were loosely linked (showed higher recombination) (Figure 4.11, Cross B). For example he found that the genes white and yellow were very tightly linked and showed only 1.3 per cent recombination while white and miniature wing showed 37.2 per cent recombination. His student Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome. Today genetic maps

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BIO12L-CH04-P64 · 4.3.3 Linkage and Recombination

> Morgan and his group knew that the genes were located on the X chromosome (Section 4.4) and saw quickly that when the two genes in a dihybrid cross were situated on the same chromosome, the proportion of parental gene combinations were much higher than the non-parental type. Morgan attributed this due to the physical association or linkage of the two genes and coined the term linkage to describe this physical association of genes on

a chromosome and the term recombination to describe the generation of non-parental gene combinations (Figure 4.11). Morgan and his group also found that even when genes were grouped on the same chromosome, some genes were very tightly linked (showed very low recombination) (Figure 4.11, Cross A) while others were loosely linked (showed higher recombination) (Figure 4.11, Cross B). For example he found that the genes white and yellow were very tightly linked and showed only 1.3 per cent recombination while white and miniature wing showed 37.2 per cent recombination. His student Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome. Today genetic maps are extensively used as a starting point in the sequencing of whole genomes as was done in the case of the Human Genome Sequencing Project, described later.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 1 FOUND

2016 Code X · Q53 In a testcross involving F_1 dihybrid flies, more parental – type offspring were produced than the recombinant – type offspring. This indicates: medium · answer 3

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
2	2 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Principles of Inheritance and Variation HIGH CONFIDENCE
WHY THIS CHAPTER	The answer is the historical attribution of the term 'linkage' to Morgan's Drosophila work, which is taught in the linkage and recombination section.
SUBTOPIC 1	Linkage & Recombination
WHY THIS SUBTOPIC	The answer is that Morgan coined the term 'linkage' while studying linked genes in Drosophila.
DIFFICULTY	**easy** · 1 step
STEPS	recall who coined the term 'linkage'
WHY	A single name-to-term recall with no reasoning attached.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`name_or_identify`

MEANS a name, term, colour, organism, scientist

E3 Content shape

what is being tested, plus the actual concept

`scientist_history`

MEANS who did it, when, or which body issued it — an attribution or a date with no subject reasoning to fall back on

> *Attribution of linkage: Morgan named it from his Drosophila crosses, while Sutton and Boveri gave the chromosomal theory of inheritance*

E4 **Thinking**

three separate questions, not one

recall

MEANS one stored fact decides it

single_step

MEANS one step, however much knowledge it needs

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **scientist_swap**

MEANS the work is attributed to the wrong person

WHY HERE Sutton is the name a student reaches for whenever the question is about genes sitting on chromosomes, because the chromosomal theory of inheritance is his. Linkage was named by Morgan, from crosses in which two genes on one chromosome failed to assort independently.

OPTION 3 **scientist_swap**

MEANS the work is attributed to the wrong person

WHY HERE Boveri's name is printed beside Sutton's in the chromosomal theory, so a student who stores the pair as one unit and reaches for the chromosome work can land on either half of it. Neither of them coined linkage; the term came out of Morgan's fly-room crosses.

OPTION 4 **scientist_swap**

MEANS the work is attributed to the wrong person

WHY HERE Linkage is taught as the exception to independent assortment, so a student who reaches for the author of the law rather than the author of the exception picks Mendel. Mendel's pea genes did assort independently; Morgan named the case where they do not.

2015 Code A **Q90** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Principles of Inheritance and Variation	Pleiotropy, Multiple Alleles & Polygenic Inheritance	easy	1

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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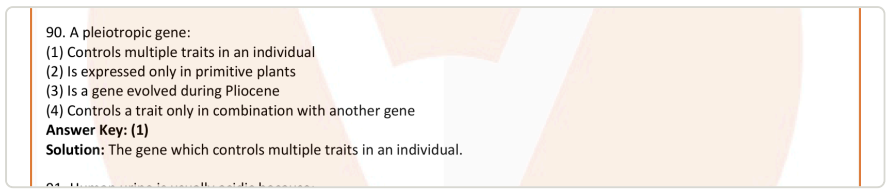
90. A pleiotropic gene:
(1) Controls multiple traits in an individual
(2) Is expressed only in primitive plants
(3) Is a gene evolved during Pliocene
(4) Controls a trait only in combination with another gene
Answer Key: (1)
Solution: The gene which controls multiple traits in an individual.

READ FROM THE IMAGE BY B3-QUESTION-READER

A pleiotropic gene:

(1) Controls multiple traits in an individual ←printed answer

(2) Is expressed only in primitive plants

(3) Is a gene evolved during Pliocene

(4) Controls a trait only in combination with another gene

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

how NCERT reaches this answer: **states answer**

4.4 Polygenic Inheritance

Mendel's studies mainly described those traits that have distinct alternate forms such as flower colour which are either purple or white. But if you look around you will find that there are many traits which are not so distinct in their occurrence and are spread across a gradient. For example, in humans we don't just have tall or short people as two distinct alternatives but a whole range of possible heights. Such traits are generally controlled by three or more genes and are thus called as polygenic traits. Besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment. Human skin colour is another classic example for this. In a polygenic trait the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive. To understand this better let us assume that three genes A, B, C control skin colour in human with the dominant forms A, B and C responsible for dark skin colour and the recessive forms a, b and c for light skin colour. The genotype with all the dominant alleles (AABBCC) will have the darkest skin colour and that with all the recessive alleles (aabbcc) will have the lightest skin colour. As expected the genotype with three dominant alleles and three recessive alleles will have an intermediate skin colour. In this manner the number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.

4.5 Pleiotropy

We have so far seen the effect of a gene on a single phenotype or trait. There are however instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes. An example of this is the disease phenylketonuria, which occurs in humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

4.6 Sex Determination

The mechanism of sex determination has always been a puzzle before the geneticists. The initial clue about the genetic/chromosomal mechanism of sex determination can be traced back to some of the experiments carried out in insects. In fact, the cytological observations made in a number of insects led to the development of the concept of genetic/chromosomal basis of sex-determination. Henking (1891) could trace a specific nuclear structure all through spermatogenesis in a few insects, and it was also observed by him that 50 per cent of the sperm received this structure after spermatogenesis, whereas the other 50 per cent sperm did not receive it. Henking gave a name to this structure as the **X body** but he could not explain its significance. Further investigations by other scientists led to the conclusion that the 'X body' of Henking was in fact a chromosome

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BIO12L-CH04-P67 · 4.5 PLEIOTROPY

4.5 PLEIOTROPY We have so far seen the effect of a gene on a single phenotype or trait. There are however instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes. An example of this is the disease phenylketonuria, which occurs in

> humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 2 FOUND

2016 Code X · Q55	Match the terms in column I with their description in Column II and choose the correct option:	medium · answer 1
2020 Code F1 · Q48	Identify the wrong statement with reference to the gene 'I' that controls ABO blood groups.	unrated

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Principles of Inheritance and Variation HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the definition of pleiotropy: a single gene producing multiple phenotypic effects, as in the phenylketonuria or starch-synthesis examples.
SUBTOPIC 1	Pleiotropy, Multiple Alleles & Polygenic Inheritance
WHY THIS SUBTOPIC	The answer depends on the definition of pleiotropy - a single gene producing multiple phenotypic effects in an individual.
DIFFICULTY	**easy** · 1 step
STEPS	recall the definition of pleiotropy - one gene affecting several phenotypic traits
WHY	A single definitional recall; the remaining options are not alternative definitions but unrelated claims.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`pick_correct_statement`

MEANS a whole claim graded as correct — a sentence, a balanced equation, a described process — or the incorrect one when the stem is negated

E3 Content shape

what is being tested, plus the actual concept

`definition_term`

MEANS what a term or quantity means, including holding two neighbouring terms apart; the strict reading of the definition decides it

> *Pleiotropy: one gene with several unrelated effects, as against several genes controlling one trait*

E4 Thinking

three separate questions, not one

concept_discrimination

MEANS choosing between two neighbouring concepts

single_step

MEANS one step, however much knowledge it needs

E5 Trap

one option, one mistake, or nothing

OPTION 4 **semantic_reversal**

MEANS the relationship is reversed in words

WHY HERE Many genes to one trait is the other way the gene-and-trait relation is taught, so a student who keeps the words one and many but not which side carries which reads this as the same claim. It is the reverse: a trait that needs a second gene is polygenic or complementary inheritance, where a pleiotropic gene has several traits following from it, as phenylketonuria brings mental retardation along with reduced hair and skin pigmentation.

OPTION 3 **look_alike_term**

MEANS a term resembling the right one

WHY HERE Pleiotropic and Pliocene carry the same pleio- / plio- sound, so a student reading the answer out of the word's √rather than its meaning is pulled to the epoch. Pleio- means more - one gene with more than one effect - and the Pliocene says nothing about how a gene acts.

2015 Code A **Q107** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Principles of Inheritance and Variation	Incomplete Dominance & Co-dominance	easy	1

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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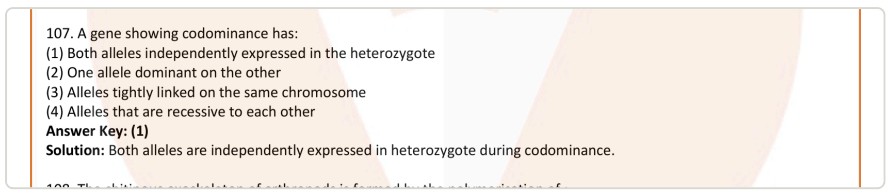

107. A gene showing codominance has:
(1) Both alleles independently expressed in the heterozygote
(2) One allele dominant on the other
(3) Alleles tightly linked on the same chromosome
(4) Alleles that are recessive to each other
Answer Key: (1)
Solution: Both alleles are independently expressed in heterozygote during codominance.

READ FROM THE IMAGE BY B3-QUESTION-READER

A gene showing codominance has:

(1) Both alleles independently expressed in the heterozygote
←printed answer

(2) One allele dominant on the other

(3) Alleles tightly linked on the same chromosome

(4) Alleles that are recessive to each other

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

how NCERT reaches this answer: **states answer**

In the first case, the modified allele is equivalent to the unmodified allele, i.e., it will produce the same phenotype/trait, i.e., result in the transformation of substrate S. Such equivalent allele pairs are very common. But, if the allele produces a non-functional enzyme or no enzyme, the phenotype may be effected. The phenotype/trait will only be dependent on the functioning of the unmodified allele. The unmodified (functioning) allele, which represents the original phenotype is the dominant allele and the modified allele is generally the recessive allele. Hence, in the example above the recessive trait is seen due to non-functional enzyme or because no enzyme is produced.

4.2.2.2 Co-dominance

Till now we were discussing crosses where the F_1 resembled either of the two parents (dominance) or was in-between (incomplete dominance). But, in the case of co-dominance the F_1 generation resembles both parents. A good example is different types of red blood cells that determine ABO blood grouping in human beings. ABO blood groups are controlled by the gene ***I***. The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene. The gene **(*I*)** has three alleles I^A, I^B and ***i***. The alleles I^A and I^B produce a slightly different form of the sugar while allele ***i*** does not produce any sugar. Because humans are diploid organisms, each person possesses any two of the three ***I*** gene alleles. I^A and I^B are completely dominant over ***i***, in other words when I^A and ***i*** are present only I^A expresses (because ***i*** does not produce any sugar), and when I^B and ***i*** are present I^B expresses. But when I^A and I^B are present together they both express their own types of sugars: this is because of co-dominance. Hence red blood cells have both A and B types of sugars. Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types (Table 4.2). *How many phenotypes are possible?*

Table 4.2: Table Showing the Genetic Basis of Blood Groups in Human Population

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood types of offspring
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	$I^A i$	A
I^B	I^A	$I^A I^B$	AB
I^B	I^B	$I^B I^B$	B
I^B	i	$I^B i$	B
i	i	$i\,i$	O

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BIO12L-CH04-P39 · 4.2.2.2 Co-dominance

4.2.2.2 Co-dominance Till now we were discussing crosses where the F1 resembled either of the two parents (dominance) or was in-between (incomplete dominance). But, in the case of co-dominance the F1 generation resembles both parents. A good example is different types of red blood cells that determine ABO blood grouping in human beings. ABO blood groups are controlled by the gene I. The plasma membrane of the red blood

> cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene. The gene (I) has three alleles I^A, I^B and i. The alleles I^A and I^B produce a slightly different form of the sugar while allele i does not produce any sugar. Because humans are diploid organisms, each person possesses any two of the three I gene alleles. I^A and I^B are completely dominant over i, in other words when I^A and i are present only I^A expresses (because i does not produce any sugar), and when I^B and i are present I^B expresses. But when I^A and I^B are present together they both express their own types of sugars: this is because of co-dominance. Hence red blood cells have both A and B types of sugars. Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types (Table 4.2). How many phenotypes are possible?

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 2 FOUND

2016 Code X · Q55	Match the terms in column I with their description in Column II and choose the correct option:	medium · answer 1
2020 Code F1 · Q48	Identify the wrong statement with reference to the gene 'I' that controls ABO blood groups.	unrated

3 The answer
the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is
subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Principles of Inheritance and Variation HIGH CONFIDENCE
WHY THIS CHAPTER	The answer is the definition of codominance - both alleles expressed independently in the heterozygote - taught as a deviation from Mendelian dominance.
SUBTOPIC 1	Incomplete Dominance & Co-dominance
WHY THIS SUBTOPIC	The answer is the definition of codominance - both alleles expressed independently and fully in the heterozygote.
DIFFICULTY	**easy** · 1 step
STEPS	recall the definition of codominance: both alleles expressed independently in the heterozygote
WHY	One definition recalled and read straight off the options.

5 How it is categorised
the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type
the shape on the page
`plain_mcq`
MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format
what the ANSWER must contain
`pick_correct_statement`
MEANS a whole claim graded as correct — a sentence, a balanced equation, a described process — or the incorrect one when the stem is negated

E3 Content shape
what is being tested, plus the actual concept
`definition_term`
MEANS what a term or quantity means, including holding two neighbouring terms apart; the strict reading of the definition decides it

> *Codominance: both alleles expressed in the heterozygote, as in the AB blood group, against complete dominance*

E4 Thinking

three separate questions, not one

`concept_discrimination`

MEANS choosing between two neighbouring concepts

`single_step`

MEANS one step, however much knowledge it needs

E5 Trap

one option, one mistake, or nothing

OPTION 2 **`other_one_fact`**

MEANS a fact about a different member of the same family — another hormone, another organelle, another syndrome

WHY HERE This is the definition of complete dominance, the member of the same family of allelic interactions that a student meets first and holds most firmly. Codominance is the case where that does not happen: neither allele masks the other and both show in the heterozygote.

OPTION 4 **`semantic_reversal`**

MEANS the relationship is reversed in words

WHY HERE A student who turns "neither allele is dominant" into "each is recessive to the other" reads this as a restatement of codominance. The words are reversed: in codominance both alleles are expressed, where mutual recessiveness would mean neither is.

OPTION 3 **`neighbouring_fact`**

MEANS a fact about an adjacent concept, disorder or event, where the families differ rather than the members

WHY HERE The prefix co- reads as "together", and linkage is the chapter's concept about things travelling together on one chromosome, so a student decoding the word instead of recalling the definition lands here. Linkage is about two different genes; the two alleles of one gene sit at the same locus on the two homologues and are never on the same chromosome.

2015 Code A **Q116** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Molecular Basis of Inheritance	The DNA	easy	4

1 The question

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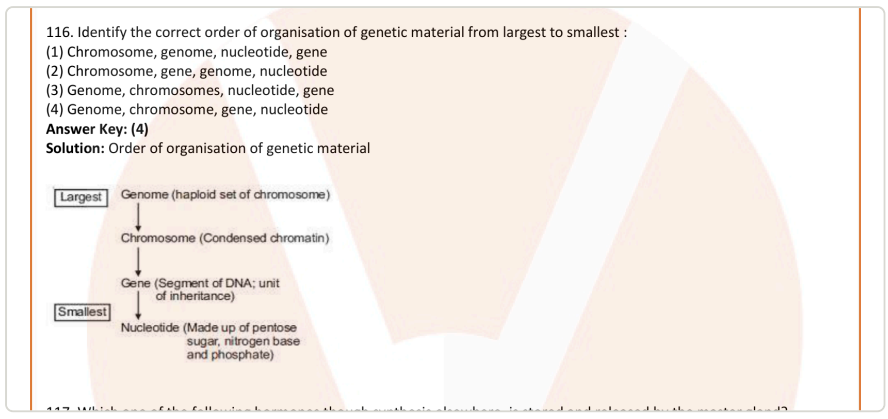
116. Identify the correct order of organisation of genetic material from largest to smallest :
(1) Chromosome, genome, nucleotide, gene
(2) Chromosome, gene, genome, nucleotide
(3) Genome, chromosomes, nucleotide, gene
(4) Genome, chromosome, gene, nucleotide
Answer Key: (4)
Solution: Order of organisation of genetic material

READ FROM THE IMAGE BY B3-QUESTION-READER

Identify the correct order of organisation of genetic material from largest to smallest :

(1) Chromosome, genome, nucleotide, gene

(2) Chromosome, gene, genome, nucleotide

(3) Genome, chromosomes, nucleotide, gene

(4) Genome, chromosome, gene, nucleotide ←printed answer

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **a genome is a set of chromosomes, a chromosome many genes, a gene many nucleotides**

APPROACH Check each option's sequence against what contains what, from the haploid chromosome set down to the single nucleotide, to find the one listing that runs consistently largest to smallest.

ALSO USES **a nucleotide is sugar + base + phosphate, the smallest unit of DNA**

how NCERT reaches this answer: **rules chained**

5.9 Human Genome Project

In the preceding sections you have learnt that it is the sequence of bases in DNA that determines the genetic information of a given organism. In other words, genetic make-up of an organism or an individual lies in the DNA sequences. If two individuals differ, then their DNA sequences should also be different, at least at some places. These assumptions led to the quest of finding out the complete DNA sequence of human genome. With the establishment of genetic engineering techniques where it was possible to isolate and clone any piece of DNA and availability of simple and fast techniques for determining DNA sequences, a very ambitious project of sequencing human genome was launched in the year 1990.

Human Genome Project (HGP) was called a mega project. You can imagine the magnitude and the requirements for the project if we simply define the aims of the project as follows:

Human genome is said to have approximately 3 x 10^9 bp, and if the cost of sequencing required is US $ 3 per bp (the estimated cost in the beginning), the total estimated cost of the project would be approximately 9 billion US dollars. Further, if the obtained sequences were to be stored in typed form in books, and if each page of the book contained 1000 letters and each book contained 1000 pages, then 3300 such books would be required to store the information of DNA sequence from a single human cell. The enormous amount of data expected to be generated also necessitated the use of high speed computational devices for data storage and retrieval, and analysis. HGP was closely associated with the rapid development of a new area in biology called **Bioinformatics.**

Goals of HGP

Some of the important goals of HGP were as follows:

(i) Identify all the approximately 20,000-25,000 genes in human DNA;
(ii) Determine the sequences of the 3 billion chemical base pairs that make up human DNA;
(iiii) Store this information in databases;
(iv) Improve tools for data analysis;
(v) Transfer related technologies to other sectors, such as industries;
(vi) Address the ethical, legal, and social issues (ELSI) that may arise from the project.

The Human Genome Project was a 13-year project coordinated by the U.S. Department of Energy and the National Institute of Health. During the early years of the HGP, the Wellcome Trust (U.K.) became a major partner; additional contributions came from Japan, France, Germany, China and others. The project was completed in 2003. Knowledge about the effects of DNA variations among individuals can lead to revolutionary new ways to diagnose, treat and someday prevent the thousands of

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5.9 HUMAN GENOME PROJECT In the preceding sections you have learnt that it is the sequence of bases in DNA that determines the genetic information of a given organism. In other words, genetic make-up of an organism or an individual lies in the DNA sequences. If two individuals differ, then their DNA sequences should also be different, at least at some places. These assumptions led to the quest of finding out the complete DNA sequence of human genome. With the establishment of genetic engineering techniques where it was possible to isolate and clone any piece of

> DNA and availability of simple and fast techniques for determining DNA sequences, a very ambitious project of sequencing human genome was launched in the year 1990.

sequenced). Another challenging task was assigning the genetic and physical maps on the genome. This was generated using information on polymorphism of restriction endonuclease recognition sites, and some repetitive DNA sequences known as microsatellites (one of the applications of polymorphism in repetitive DNA sequences shall be explained in next section of DNA fingerprinting).

5.9.1 Salient Features of Human Genome

Some of the salient observations drawn from human genome project are as follows:

(i) The human genome contains 3164.7 million bp.

(ii) The average gene consists of 3000 bases, but sizes vary greatly, with the largest known human gene being dystrophin at 2.4 million bases.

(iii) The total number of genes is estimated at 30,000–much lower than previous estimates of 80,000 to 1,40,000 genes. Almost all (99.9 per cent) nucleotide bases are exactly the same in all humans.

(iv) The functions are unknown for over 50 per cent of the discovered genes.

(v) Less than 2 per cent of the genome codes for proteins.

(vi) Repeated sequences make up very large portion of the human genome.

(vii) Repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundred to thousand times. They are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution.

(viii) Chromosome 1 has most genes (2968), and the Y has the fewest (231).

(ix) Scientists have identified about 1.4 million locations where single-base DNA differences (**SNPs – single nucleotide polymorphism**, pronounced as 'snips') occur in humans. This information promises to revolutionise the processes of finding chromosomal locations for disease-associated sequences and tracing human history.

5.9.2 Applications and Future Challenges

Deriving meaningful knowledge from the DNA sequences will define research through the coming decades leading to our understanding of biological systems. This enormous task will require the expertise and creativity of tens of thousands of scientists from varied disciplines in both the public and private sectors worldwide. One of the greatest impacts of having the HG sequence may well be enabling a radically new approach to biological research. In the past, researchers studied one or a few genes at a time. With whole-genome sequences and new high-throughput technologies, we can approach questions systematically and on a much

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BIO12L-CH05-P134 · 5.9.1 Salient Features of Human Genome

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In some viruses the flow of information is in reverse direction, that is, from RNA to DNA. *Can you suggest a simple name to the process?*

5.1.2 Packaging of DNA Helix

Taken the distance between two consecutive base pairs as 0.34 nm (0.34×10^{-9} m), if the length of DNA double helix in a typical mammalian cell is calculated (simply by multiplying the total number of bp with distance between two consecutive bp, that is, 6.6×10^{9} bp × 0.34×10^{-9}m/bp), it comes out to be approximately 2.2 metres. A length that is far greater than the dimension of a typical nucleus (approximately 10^{-6} m). How is such a long polymer packaged in a cell?

If the length of E. coli DNA is 1.36 mm, can you calculate the number of base pairs in E.coli?

In prokaryotes, such as, *E. coli*, though they do not have a defined nucleus, the DNA is not scattered throughout the cell. DNA (being negatively charged) is held with some proteins (that have positive charges) in a region termed as 'nucleoid'. The DNA in nucleoid is organised in large loops held by proteins.

In eukaryotes, this organisation is much more complex. There is a set of positively charged, basic proteins called **histones**. A protein acquires charge depending upon the abundance of amino acids residues with charged side chains. Histones are rich in the basic amino acid residues lysine and arginine. Both the amino acid residues carry positive charges in their side chains. Histones are organised to form a unit of eight molecules called **histone octamer**. The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called **nucleosome** (Figure 5.4 a). A typical nucleosome contains 200 bp of DNA helix. Nucleosomes constitute the repeating unit of a structure in nucleus called **chromatin,** thread-like stained (coloured) bodies seen in nucleus. The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope (EM) (Figure 5.4 b).

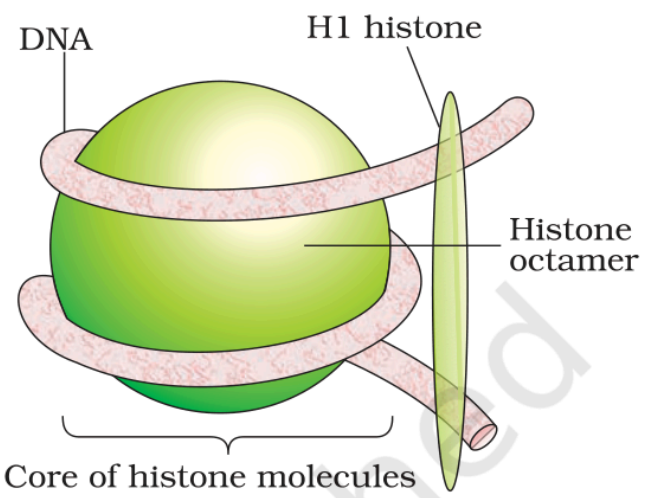

Figure 5.4a Nucleosome

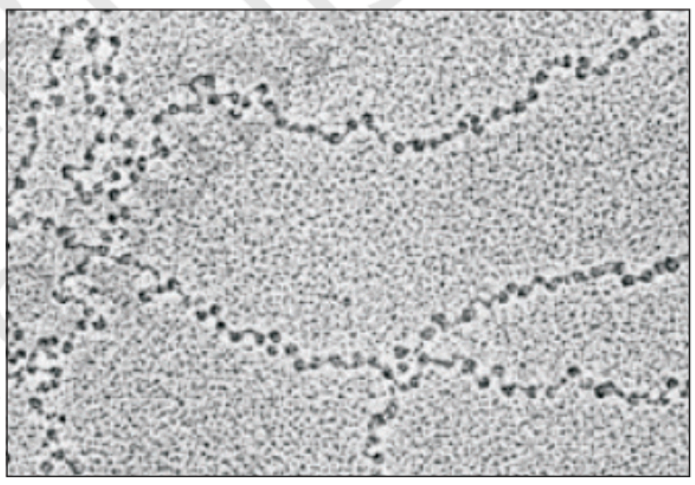

Figure 5.4b EM picture - 'Beads-on-String'

Theoretically, how many such beads (nucleosomes) do you imagine are present in a mammalian cell?

The beads-on-string structure in chromatin is packaged to form chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form chromosomes. The packaging of chromatin at higher level requires additional set of proteins that collectively are referred to as

BIO12L-CH05-P36 · 5.1.2 Packaging of DNA Helix

> The beads-on-string structure in chromatin is packaged to form chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form chromosomes. The packaging of chromatin at higher level requires additional set of proteins that collectively are referred to as Non-histone Chromosomal (NHC) proteins. In a typical nucleus, some region of chromatin are loosely packed (and stains light) and are referred to as euchromatin. The chromatin that is more densely packed and stains dark are called as Heterochromatin. Euchromatin is said to be transcriptionally active chromatin, whereas heterochromatin is inactive.

There is a convention in defining the two strands of the DNA in the structural gene of a transcription unit. Since the two strands have opposite polarity and the **DNA-dependent RNA polymerase** also catalyse the polymerisation in only one direction, that is, 5′–3′, the strand that has the polarity 3′–5′ acts as a template, and is also referred to as **template strand**. The other strand which has the polarity (5′–3′) and the sequence same as RNA (except thymine at the place of uracil), is displaced during transcription. Strangely, this strand (which does not code for anything) is referred to as **coding strand**. All the reference point while defining a transcription unit is made with coding strand. To explain the point, a hypothetical sequence from a transcription unit is represented below:

3′-ATGCATGCATGCATGCATGCATGC-5′ Template Strand

5′-TACGTACGTACGTACGTACGTACG-3′ Coding Strand

Can you now write the sequence of RNA transcribed from the above DNA?

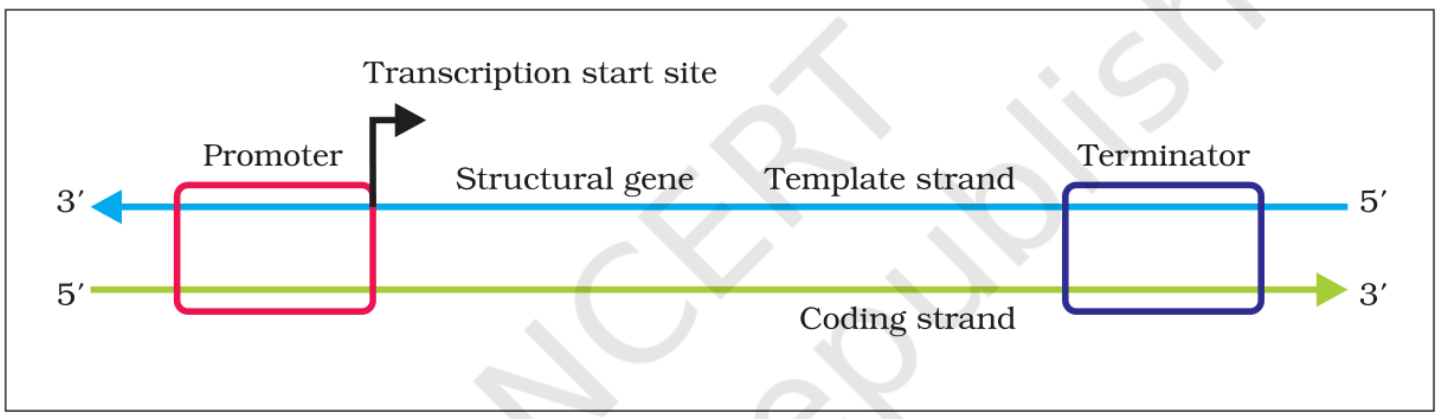

Figure 5.9 Schematic structure of a transcription unit

The **promoter** and **terminator** flank the **structural gene** in a transcription unit. The promoter is said to be located towards 5′-end (upstream) of the structural gene (the reference is made with respect to the polarity of coding strand). It is a DNA sequence that provides binding site for RNA polymerase, and it is the presence of a promoter in a transcription unit that also defines the template and coding strands. By switching its position with terminator, the definition of coding and template strands could be reversed. The terminator is located towards 3′-end (downstream) of the coding strand and it usually defines the end of the process of transcription (Figure 5.9). There are additional regulatory sequences that may be present further upstream or downstream to the promoter. Some of the properties of these sequences shall be discussed while dealing with regulation of gene expression.

5.5.2 Transcription Unit and the Gene

A gene is defined as the functional unit of inheritance. Though there is no ambiguity that the genes are located on the DNA, it is difficult to literally

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BIO12L-CH05-P84 · 5.5.2 Transcription Unit and the Gene

> 5.5.2 Transcription Unit and the Gene A gene is defined as the functional unit of inheritance. Though there is no ambiguity that the genes are located on the DNA, it is difficult to literally define a gene in terms of DNA sequence. The DNA sequence coding for tRNA or rRNA molecule also define a gene. However by defining a cistron as a segment of DNA coding for a polypeptide, the structural gene in a transcription unit could be said as monocistronic (mostly in eukaryotes) or polycistronic (mostly in bacteria or prokaryotes). In eukaryotes, the monocistronic structural genes have interrupted coding sequences – the genes in eukaryotes are split. The coding sequences or expressed sequences are defined as exons. Exons are said

> to be those sequence that appear in mature or processed RNA. The exons are interrupted by introns. Introns or intervening sequences do not appear in mature or processed RNA. The split-gene arrangement further complicates the definition of a gene in terms of a DNA segment.

of complete nucleotide sequence of human genome during last decade has set in a new era of genomics. In the last section, the essentials of human genome sequencing and its consequences will also be discussed.

Let us begin our discussion by first understanding the structure of the most interesting molecule in the living system, that is, the DNA. In subsequent sections, we will understand that why it is the most abundant genetic material, and what its relationship is with RNA.

5.1 The DNA

DNA is a long polymer of deoxyribonucleotides. The length of DNA is usually defined as number of nucleotides (or a pair of nucleotide referred to as base pairs) present in it. This also is the characteristic of an organism. For example, a bacteriophage known as ϕ×174 has 5386 nucleotides, Bacteriophage lambda has 48502 base pairs (bp), *Escherichia coli* has 4.6×10^6 bp, and haploid content of human DNA is 3.3×10^9 bp. Let us discuss the structure of such a long polymer.

5.1.1 Structure of Polynucleotide Chain

Let us recapitulate the chemical structure of a polynucleotide chain (DNA or RNA). A nucleotide has three components – a nitrogenous base, a pentose sugar (ribose in case of RNA, and deoxyribose for DNA), and a phosphate group. There are two types of nitrogenous bases – Purines (Adenine and Guanine), and Pyrimidines (Cytosine, Uracil and Thymine). Cytosine is common for both DNA and RNA and Thymine is present in DNA. Uracil is present in RNA at the place of Thymine. A nitrogenous base is linked to the OH of 1′C pentose sugar through a N-glycosidic linkage to form a nucleoside, such as adenosine or deoxyadenosine, guanosine or deoxyguanosine, cytidine or deoxycytidine and uridine or deoxythymidine. When a phosphate group is linked to OH of 5′C of a nucleoside through phosphoester linkage, a corresponding nucleotide (or deoxynucleotide depending upon the type of sugar present) is formed. Two nucleotides are linked through 3′-5′ phosphodiester linkage to form a dinucleotide. More nucleotides can be joined in such a manner to form a polynucleotide chain. A polymer thus formed has at one end a free

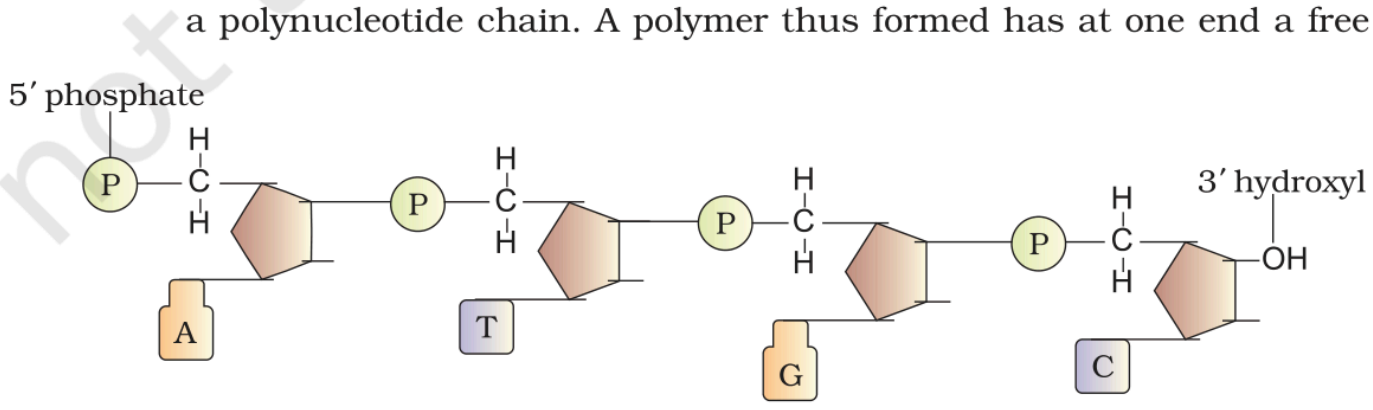

Figure 5.1 A Polynucleotide chain

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5.1 THE DNA DNA is a long polymer of deoxyribonucleotides. The length of DNA is usually defined as number of nucleotides (or a pair of nucleotide referred to as base pairs) present in it. This also is the characteristic of an organism. For example, a bacteriophage known as φ×174 has 5386 nucleotides, Bacteriophage lambda has 48502 base pairs (bp), Escherichia coli has 4.6 × 106 bp, and haploid content of human DNA is 3.3 × 109 bp. Let us discuss the structure of such a long polymer.

HOW THE QUOTED LINES REACH THIS ANSWER

1. Genome is the complete DNA of the organism and measures 3164.7 million bp, so it is the largest level.
2. A chromosome is one packaged DNA molecule, and the genome is made of many of them (Chromosome 1 and the Y are two of the set), so genome > chromosome.
3. One chromosome carries thousands of genes (2968 on Chromosome 1), so chromosome > gene.
4. One gene averages 3000 bases, i.e. thousands of nucleotides, and the nucleotide is the monomer DNA is polymerised from, so gene > nucleotide.
5. Order from largest to smallest = genome, chromosome, gene, nucleotide, which is option 4.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
4	4 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Molecular Basis of Inheritance HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the hierarchy of genetic material — genome contains chromosomes, a chromosome carries genes, and a gene is a run of nucleotides.
SUBTOPIC 1	The DNA
WHY THIS SUBTOPIC	The answer depends on the size hierarchy of genetic material — genome, then chromosome, then gene, then nucleotide as the monomer.
DIFFICULTY	**easy** · 1 step
STEPS	order the four terms by what contains what: genome, then chromosome, then gene, then nucleotide
WHY	One ordering that follows directly from the definitions of the four terms.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`arrange_in_order`

MEANS items to sequence, answer is an order

E2 Format

what the ANSWER must contain

`pick_order`

MEANS an ordering of items

E3 Content shape

what is being tested, plus the actual concept

`definition_term`

MEANS what a term or quantity means, including holding two neighbouring terms apart; the strict reading of the definition decides it

> *Levels of genetic material: the genome is all the chromosomes, a chromosome carries many genes, and a gene is a run of nucleotides*

E4 **Thinking**

three separate questions, not one

`application`

MEANS a stated rule, criterion or convention applied to this case. The rule must be known first — application is not knowledge-free

`single_step`

MEANS one step, however much knowledge it needs

`elimination`

MEANS name the one pair or fact and the options it

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **`semantic_reversal`**

MEANS the relationship is reversed in words

WHY HERE The chromosome is the thing a student can picture down a microscope while the genome is only a word, so a student ranking by how concrete each term feels puts the chromosome on top and the genome inside it. The genome is the whole set of chromosomes, so the nesting runs the other way.

OPTION 2 **`no_derivation_route`**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE The two mis-steps that do reach printed options are starting the ladder at the chromosome and swapping gene with nucleotide. Neither puts the genome third: it contains every gene and every nucleotide, so no reading of the four terms lands it between them.

OPTION 3 **`sequence_endpoint_or_adjacency`**

MEANS an ordering that gets the ends right and fails on one interior adjacency, or reproduces the printed list order

WHY HERE A student who settles the top of the ladder and then ranks the last two by how concrete they sound - a nucleotide is a molecule you can draw, a gene only an instruction - puts the nucleotide above the gene. A gene is a stretch of many nucleotides, so the gene is the larger of the two and that last adjacency is the only thing this option gets wrong.

2015 Code A **Q120** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Molecular Basis of Inheritance	Salient Features of Structure of DNA	easy	1

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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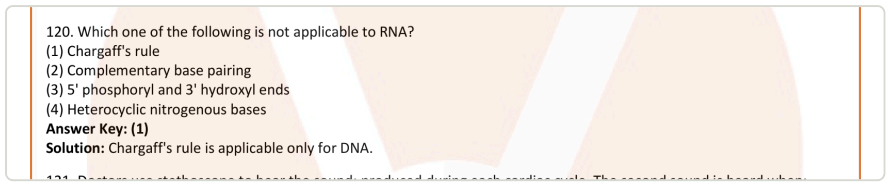
120. Which one of the following is not applicable to RNA?
(1) Chargaff's rule
(2) Complementary base pairing
(3) 5' phosphoryl and 3' hydroxyl ends
(4) Heterocyclic nitrogenous bases
Answer Key: (1)
Solution: Chargaff's rule is applicable only for DNA.

READ FROM THE IMAGE BY B3-QUESTION-READER

Which one of the following is not applicable to RNA?

(1) Chargaff's rule ←printed answer

(2) Complementary base pairing

(3) 5' phosphoryl and 3' hydroxyl ends

(4) Heterocyclic nitrogenous bases

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

how NCERT reaches this answer: **concept applied**

phosphate moiety at 5′-end of sugar, which is referred to as 5'-end of polynucleotide chain. Similarly, at the other end of the polymer the sugar has a free OH of 3′C group which is referred to as 3′-end of the polynucleotide chain. The backbone of a polynucleotide chain is formed due to sugar and phosphates. The nitrogenous bases linked to sugar moiety project from the backbone (Figure 5.1).

In RNA, every nucleotide residue has an additional –OH group present at 2′-position in the ribose. Also, in RNA the uracil is found at the place of thymine (5-methyl uracil, another chemical name for thymine).

DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869. He named it as 'Nuclein'. However, due to technical limitation in isolating such a long polymer intact, the elucidation of structure of DNA remained elusive for a very long period of time. It was only in 1953 that James Watson and Francis Crick, based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin, proposed a very simple but famous **Double Helix** model for the structure of DNA. One of the hallmarks of their proposition was base pairing between the two strands of polynucleotide chains. However, this proposition was also based on the observation of Erwin Chargaff that for a double stranded DNA, the ratios between **Adenine** and **Thymine** and **Guanine** and **Cytosine** are constant and equals one.

The base pairing confers a very unique property to the polynucleotide chains. They are said to be complementary to each other, and therefore if the sequence of bases in one strand is known then the sequence in other strand can be predicted. Also, if each strand from a DNA (let us call it as a parental DNA) acts as a template for synthesis of a new strand, the two double stranded DNA (let us call them as daughter DNA) thus, produced would be identical to the parental DNA molecule. Because of this, the genetic implications of the structure of DNA became very clear.

The salient features of the Double-helix structure of DNA are as follows:

(i) It is made of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases project inside.

(ii) The two chains have anti-parallel polarity. It means, if one chain has the polarity 5′→3′, the other has 3′→5′.

(iii) The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds. As a result, always a purine comes opposite to a pyrimidine. This generates approximately uniform distance between the two strands of the helix (Figure 5.2).

(iv) The two chains are coiled in a right-handed fashion. The pitch of the helix is 3.4 nm (a nanometre is one billionth of a metre, that is 10^{-9} m) and there are roughly 10 bp in each

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BIO12L-CH05-P20 · 5.1.1 Structure of Polynucleotide Chain

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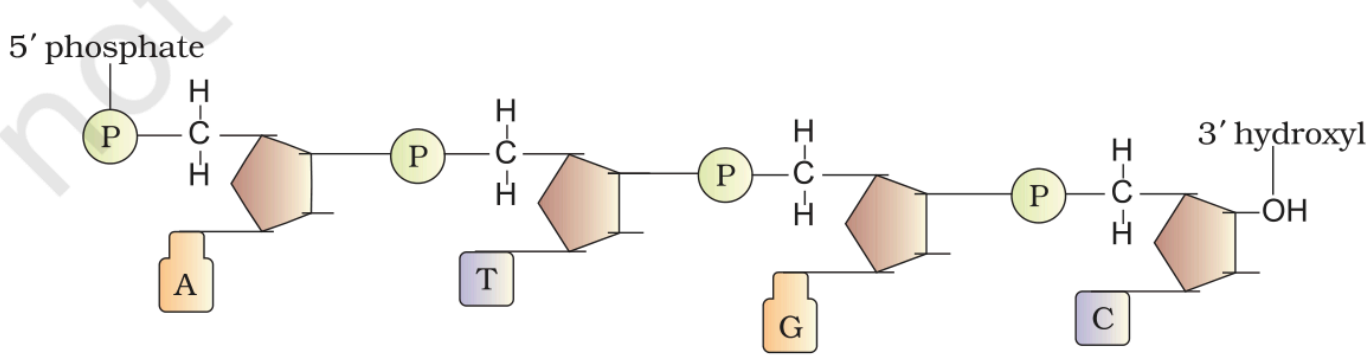

Figure 5.1 A Polynucleotide chain

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and uridine or deoxythymidine. When a phosphate group is linked to OH of 5' C of a nucleoside through phosphoester linkage, a corresponding nucleotide (or deoxynucleotide depending upon the type of sugar present) is formed. Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide. More nucleotides can be joined in such a manner to form a polynucleotide chain. A polymer thus formed has at one end a free phosphate moiety at 5'-end of sugar, which is referred to as 5'-end of polynucleotide chain. Similarly, at the other end of the polymer the sugar has a free OH of 3'C group which is referred to as 3' -end of the polynucleotide chain. The backbone of a polynucleotide chain is formed due to sugar and phosphates. The nitrogenous bases linked to sugar moiety project from the backbone (Figure 5.1).

in some viruses, RNA is the genetic material (for example, Tobacco Mosaic viruses, QB bacteriophage, etc.). Answer to some of the questions such as, why DNA is the predominant genetic material, whereas RNA performs dynamic functions of messenger and adapter has to be found from the differences between chemical structures of the two nucleic acid molecules.

Can you recall the two chemical differences between DNA and RNA?

A molecule that can act as a genetic material must fulfill the following criteria:

(i) It should be able to generate its replica (Replication).

(ii) It should be stable chemically and structurally.

(iii) It should provide the scope for slow changes (mutation) that are required for evolution.

(iv) It should be able to express itself in the form of 'Mendelian Characters'.

If one examines each requirement one by one, because of rule of base pairing and complementarity, both the nucleic acids (DNA and RNA) have the ability to direct their duplications. The other molecules in the living system, such as proteins fail to fulfill first criteria itself.

The genetic material should be stable enough not to change with different stages of life cycle, age or with change in physiology of the organism. Stability as one of the properties of genetic material was very evident in Griffith's 'transforming principle' itself that heat, which killed the bacteria, at least did not destroy some of the properties of genetic material. This now can easily be explained in light of the DNA that the two strands being complementary if separated by heating come together, when appropriate conditions are provided. Further, 2′-OH group present at every nucleotide in RNA is a reactive group and makes RNA labile and easily degradable. RNA is also now known to be catalytic, hence reactive. Therefore, DNA chemically is less reactive and structurally more stable when compared to RNA. Therefore, among the two nucleic acids, the DNA is a better genetic material.

In fact, the presence of thymine at the place of uracil also confers additional stability to DNA. (Detailed discussion about this requires understanding of the process of repair in DNA, and you will study these processes in higher classes.)

Both DNA and RNA are able to mutate. In fact, RNA being unstable, mutate at a faster rate. Consequently, viruses having RNA genome and having shorter life span mutate and evolve faster.

RNA can directly code for the synthesis of proteins, hence can easily express the characters. DNA, however, is dependent on RNA for synthesis of proteins. The protein synthesising machinery has evolved around RNA. The above discussion indicate that both RNA and DNA can function as

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BIO12L-CH05-P57 · 5.2.2 Properties of Genetic Material (DNA versus RNA)

> If one examines each requirement one by one, because of rule of base pairing and complementarity, both the nucleic acids (DNA and RNA) have the ability to direct their duplications. The other molecules in the living system, such as proteins fail to fulfill first criteria itself.

phosphate moiety at 5′-end of sugar, which is referred to as 5'-end of polynucleotide chain. Similarly, at the other end of the polymer the sugar has a free OH of 3′C group which is referred to as 3′-end of the polynucleotide chain. The backbone of a polynucleotide chain is formed due to sugar and phosphates. The nitrogenous bases linked to sugar moiety project from the backbone (Figure 5.1).

In RNA, every nucleotide residue has an additional –OH group present at 2′-position in the ribose. Also, in RNA the uracil is found at the place of thymine (5-methyl uracil, another chemical name for thymine).

DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869. He named it as 'Nuclein'. However, due to technical limitation in isolating such a long polymer intact, the elucidation of structure of DNA remained elusive for a very long period of time. It was only in 1953 that James Watson and Francis Crick, based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin, proposed a very simple but famous **Double Helix** model for the structure of DNA. One of the hallmarks of their proposition was base pairing between the two strands of polynucleotide chains. However, this proposition was also based on the observation of Erwin Chargaff that for a double stranded DNA, the ratios between **Adenine** and **Thymine** and **Guanine** and **Cytosine** are constant and equals one.

The base pairing confers a very unique property to the polynucleotide chains. They are said to be complementary to each other, and therefore if the sequence of bases in one strand is known then the sequence in other strand can be predicted. Also, if each strand from a DNA (let us call it as a parental DNA) acts as a template for synthesis of a new strand, the two double stranded DNA (let us call them as daughter DNA) thus, produced would be identical to the parental DNA molecule. Because of this, the genetic implications of the structure of DNA became very clear.

The salient features of the Double-helix structure of DNA are as follows:

(i) It is made of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases project inside.

(ii) The two chains have anti-parallel polarity. It means, if one chain has the polarity 5′→3′, the other has 3′→5′.

(iii) The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds. As a result, always a purine comes opposite to a pyrimidine. This generates approximately uniform distance between the two strands of the helix (Figure 5.2).

(iv) The two chains are coiled in a right-handed fashion. The pitch of the helix is 3.4 nm (a nanometre is one billionth of a metre, that is 10^{-9} m) and there are roughly 10 bp in each

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BIO12L-CH05-P19 · 5.1.1 Structure of Polynucleotide Chain

> In RNA, every nucleotide residue has an additional –OH group present at 2'-position in the ribose. Also, in RNA the uracil is found at the place of thymine (5-methyl uracil, another chemical name for thymine).

HOW THE QUOTED LINES REACH THIS ANSWER

1. NCERT states Chargaff's equivalence only for a double stranded DNA: the Adenine:Thymine and Guanine:Cytosine ratios are constant and equal one.
2. RNA carries uracil in place of thymine, so an Adenine-to-Thymine ratio is not even defined for RNA, and RNA is not the two-chain base-paired structure the rule is stated for.
3. Therefore Chargaff's rule is the one property of the four that does not apply to RNA, while the other three are stated for a polynucleotide chain of either kind: option 1.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 4 FOUND

2020 Code F1 · Q40	Which of the following statements is correct?	unrated
2020 Code F1 · Q74	If the distance between two consecutive base pairs is 0.34 nm and the total number of base pair os a DNA double helix in a typical mammalian cell is 6.6 X 109 bp, then the length of the DNA ...	unrated
2021 Code O1 · Q172	If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it?	unrated · answer 1
2022 Code Q1 · Q153	If the length of a DNA molecule is 1.1 metres, what will be the approximate number of base pairs?	unrated · answer 1

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Molecular Basis of Inheritance HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on Chargaff's equivalence rule being a property of double-stranded DNA and therefore not applicable to single-stranded RNA.
SUBTOPIC 1	Salient Features of Structure of DNA
SUBTOPIC 2	DNA vs RNA & Genetic Material is DNA
WHY THIS SUBTOPIC	The answer depends on Chargaff's equivalence rule being a consequence of double-stranded base pairing in DNA and therefore inapplicable to single-stranded RNA.
DIFFICULTY	**easy** · 1 step
STEPS	recall that Chargaff's equivalence rule is a statement about double-stranded DNA, so it does not hold for RNA
WHY	One recall of the scope of Chargaff's rule; the other three properties are shared by both nucleic acids.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`name_or_identify`

MEANS a name, term, colour, organism, scientist

E3 Content shape

what is being tested, plus the actual concept

`structure_chemical`

MEANS the structure and bonding of a named species: geometry, electron counts (octet, VSEPR, MO filling, π electrons), oxidation states, or a structure rebuilt from memory

> *RNA is single-stranded, so Chargaff's equivalence does not hold for it, although base pairing, 5' and 3' ends and nitrogenous bases all do*

E4 Thinking

three separate questions, not one

`concept_discrimination`

MEANS choosing between two neighbouring concepts

`several_independent_checks`

MEANS each item verified separately, verdicts combined

E5 Trap

one option, one mistake, or nothing

OPTION 2 **`true_statement_trap`**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE A student who remembers "RNA is single-stranded" concludes that base pairing cannot apply to it either, and selects it under the stem's "not". A single strand still folds back and pairs with itself, as in the cloverleaf arms of tRNA; what single-strandedness actually kills is Chargaff's A = T, G = C equivalence.

OPTION 3 **`true_statement_trap`**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE The 5' phosphoryl and 3' hydroxyl ends are labelled on the DNA polynucleotide drawn in the chapter, so a student who learned the two ends from that figure reads them as DNA's alone. RNA is a polynucleotide built the same way and carries the same two ends, so the claim is true of it and fails the stem's "not".

OPTION 4 **`true_statement_trap`**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE Uracil replaces thymine in RNA, so a student who checks the bases, finds one of the four changed, and reads "heterocyclic nitrogenous bases" as the DNA set picks this. Uracil is a pyrimidine like thymine, so all four RNA bases are heterocyclic and nitrogenous and the claim holds.

2015 Code A **Q127** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Molecular Basis of Inheritance	DNA Fingerprinting	medium	3

1 The question

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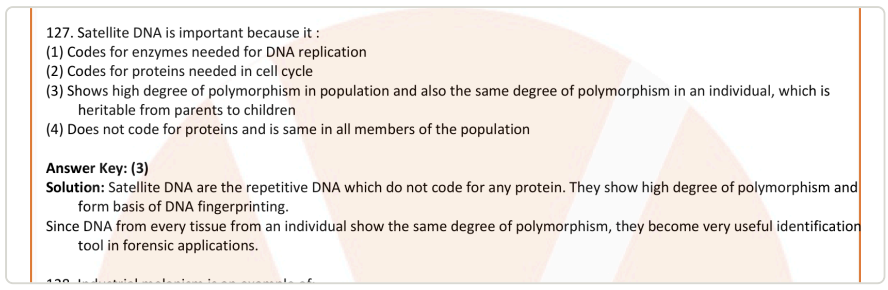

127. Satellite DNA is important because it :
(1) Codes for enzymes needed for DNA replication
(2) Codes for proteins needed in cell cycle
(3) Shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children
(4) Does not code for proteins and is same in all members of the population

Answer Key: (3)
Solution: Satellite DNA are the repetitive DNA which do not code for any protein. They show high degree of polymorphism and form basis of DNA fingerprinting.
Since DNA from every tissue from an individual show the same degree of polymorphism, they become very useful identification tool in forensic applications.

READ FROM THE IMAGE BY B3-QUESTION-READER

Satellite DNA is important because it :

(1) Codes for enzymes needed for DNA replication

(2) Codes for proteins needed in cell cycle

(3) Shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children ←printed answer

(4) Does not code for proteins and is same in all members of the population

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

how NCERT reaches this answer: **states answer**

broader scale. They can study all the genes in a genome, for example, all the transcripts in a particular tissue or organ or tumor, or how tens of thousands of genes and proteins work together in interconnected networks to orchestrate the chemistry of life.

5.10 DNA Fingerprinting

As stated in the preceding section, 99.9 per cent of base sequence among humans is the same. *Assuming human genome as* 3×10^9 *bp, in how many base sequences would there be differences?* It is these differences in sequence of DNA which make every individual unique in their phenotypic appearance. If one aims to find out genetic differences between two individuals or among individuals of a population, sequencing the DNA every time would be a daunting and expensive task. Imagine trying to compare two sets of 3×10^6 base pairs. DNA fingerprinting is a very quick way to compare the DNA sequences of any two individuals.

DNA fingerprinting involves identifying differences in some specific regions in DNA sequence called as **repetitive DNA**, because in these sequences, a small stretch of DNA is repeated many times. These repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation. The bulk DNA forms a major peak and the other small peaks are referred to as **satellite DNA**. Depending on base composition (A : T rich or G:C rich), length of segment, and number of repetitive units, the satellite DNA is classified into many categories, such as micro-satellites, mini-satellites etc. These sequences normally do not code for any proteins, but they form a large portion of human genome. These sequence show high degree of polymorphism and form the basis of DNA fingerprinting. Since DNA from every tissue (such as blood, hair-follicle, skin, bone, saliva, sperm etc.), from an individual show the same degree of polymorphism, they become very useful identification tool in forensic applications. Further, as the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing, in case of disputes.

As polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting, it is essential that we understand what DNA polymorphism means in simple terms. **Polymorphism** (variation at genetic level) arises due to mutations. (Recall different kind of mutations and their effects that you have already studied in Chapter 4, and in the preceding sections in this chapter.) New mutations may arise in an individual either in somatic cells or in the germ cells (cells that generate gametes in sexually reproducing organisms). If a germ cell mutation does not seriously impair individual's ability to have offspring who can transmit the mutation, it can spread to

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BIO12L-CH05-P137 · 5.10 DNA FINGERPRINTING

> DNA fingerprinting involves identifying differences in some specific regions in DNA sequence called as repetitive DNA, because in these sequences, a small stretch of DNA is repeated many times. These repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation. The bulk DNA forms a major peak and the other small peaks are referred to as satellite DNA. Depending on base composition (A : T rich or G:C rich), length of segment, and number of repetitive units, the satellite DNA is classified into many categories, such as micro-satellites, mini-satellites etc. These sequences normally do not code for any proteins, but they form a large portion of human genome. These sequence show

> high degree of polymorphism and form the basis of DNA fingerprinting. Since DNA from every tissue (such as blood, hair-follicle, skin, bone, saliva, sperm etc.), from an individual show the same degree of polymorphism, they become very useful identification tool in forensic applications. Further, as the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing, in case of disputes.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 3 FOUND

2016 Code X · Q126	Which of the following is not required for any of the techniques of DNA fingerprinting available at present ?	medium · answer 1
2021 Code O1 · Q144	DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as:	unrated · answer 4
2022 Code Q1 · Q117	DNA polymorphism forms the basis of :	unrated · answer 3

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Molecular Basis of Inheritance HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on satellite DNA repeats being highly polymorphic between individuals yet identical in every tissue and heritable, the basis of DNA fingerprinting.
SUBTOPIC 1	DNA Fingerprinting
WHY THIS SUBTOPIC	The answer depends on satellite/repetitive DNA showing high polymorphism between individuals while being identical in all tissues of one individual and heritable, which is the basis of DNA fingerprinting.
DIFFICULTY	**medium** · 2 steps
STEPS	recall that satellite DNA is non-coding repetitive DNA showing a high degree of population-level polymorphism that is inherited → separate that from option 4, which also calls it non-coding but asserts it is identical in all members of the population
WHY	The work is one recall about satellite DNA plus the discrimination between two options that agree on its non-coding nature and differ only on polymorphism.
THE CATCH	Option 4 is half-true, so a candidate who remembers only 'satellite DNA does not code for protein' has no basis to reject it.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`pick_correct_statement`

MEANS a whole claim graded as correct — a sentence, a balanced equation, a described process — or the incorrect one when the stem is negated

E3 Content shape

what is being tested, plus the actual concept

`definition_term`

MEANS what a term or quantity means, including holding two neighbouring terms apart; the strict reading of the definition decides it

> *Satellite DNA: non-coding repetitive sequences whose high polymorphism between individuals, inherited from the parents, is what DNA fingerprinting reads*

E4 Thinking

three separate questions, not one

recall

MEANS one stored fact decides it

several_independent_checks

MEANS each item verified separately, verdicts combined

E5 Trap

one option, one mistake, or nothing

OPTION 4 **half_true**

MEANS an option with two claims, one right

WHY HERE A student who holds satellite DNA as the non-coding repetitive fraction reads the first half, finds it exactly right, and takes the option on the strength of it. The second half is the claim that fails: the number of repeats differs so much from person to person that the variation is precisely what fingerprinting measures, so it is not the same in all members of the population.

OPTION 1 **no_derivation_route**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE The only reach available here is the general rule that DNA codes for protein, and that rule says nothing about which protein, so it cannot pick the replication enzymes of this option over the cell-cycle proteins of option 2. The coding genes lie in the non-repetitive fraction, so no fact about satellite DNA reaches either.

OPTION 2 **no_derivation_route**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE The same demonstration read the other way round: nothing a student knows about satellite DNA chooses cell-cycle proteins over replication enzymes, so options 1 and 2 stand or fall together and neither is selected by any route. Both are there to make a coding role look like one of the live answers.

2015 Code A **Q136** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Modern Physics	Spectral Lines of Hydrogen Atom	hard	1

1 The question

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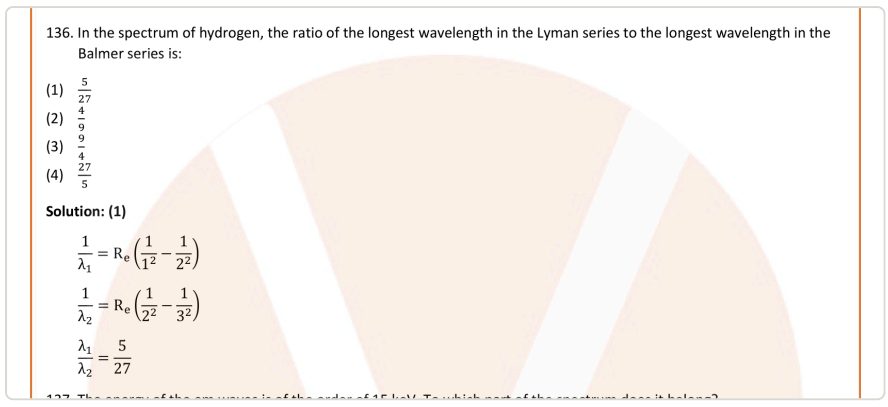

136. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is:

(1) $\frac{5}{27}$
(2) $\frac{4}{9}$
(3) $\frac{9}{4}$
(4) $\frac{27}{5}$

Solution: (1)

$$\frac{1}{\lambda_1} = R_e\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$$

$$\frac{1}{\lambda_2} = R_e\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\frac{\lambda_1}{\lambda_2} = \frac{5}{27}$$

READ FROM THE IMAGE BY B3-QUESTION-READER

In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is:

(1) $\frac{5}{27}$ ←printed answer

(2) $\frac{4}{9}$

(3) $\frac{9}{4}$

(4) $\frac{27}{5}$

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **1/lambda = R(1/n_1^2 - 1/n_2^2)**

APPROACH The longest wavelength of a series is its smallest energy jump, so take n = 2 -> 1 for Lyman and n = 3 -> 2 for Balmer and divide the two Rydberg expressions to get the ratio.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT Physics

CHAPTER Modern Physics HIGH CONFIDENCE

WHY THIS CHAPTER The answer depends on the Rydberg/Bohr formula for the hydrogen spectral series, giving the longest Lyman (2->1) and longest Balmer (3->2) wavelengths.

SUBTOPIC 1 Spectral Lines of Hydrogen Atom

WHY THIS SUBTOPIC The answer depends on the series formulae for hydrogen: the longest Lyman line is the 2->1 transition and the longest Balmer line the 3->2 transition, whose wavelength ratio is 5/27.

DIFFICULTY **hard** · 4 steps

STEPS see that the longest wavelength in a series is its smallest energy jump, so Lyman means n = 2 to 1 and Balmer means n = 3 to 2 → apply the Rydberg formula to the Lyman transition: 1/lambda = R(1 - 1/4) = 3R/4 → apply it to the Balmer transition: 1/lambda = R(1/4 - 1/9) = 5R/36 → divide the two wavelengths and reduce the ratio

WHY Two separate Rydberg evaluations preceded by the choice of which transition each 'longest wavelength' refers to, then a ratio of reciprocals.

THE CATCH The inverted ratio appears as an option, so taking the reciprocal in the wrong order lands exactly on a printed choice rather than on a wrong number.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`pick_ratio`

MEANS a ratio, as a ratio

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Rydberg formula for the longest line of a series: the longest wavelength is the smallest jump,*
>
> $n = 2 \to 1$
>
> *for Lyman and*
>
> $n = 3 \to 2$
>
> *for Balmer*

E4 Thinking

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`elimination`

MEANS name the one pair or fact and the options it

`estimation`

E5 Trap

one option, one mistake, or nothing

OPTION 4 **`inverted_ratio`**

MEANS the reciprocal of the right answer is printed

WHY HERE The Rydberg formula gives $1/\lambda$ and not λ, so a student who divides the two bracket values as they stand, $\frac{3/4}{5/36}$, gets 27/5. That is the ratio of the two photon energies - the exact reciprocal of the ratio of wavelengths the stem asks for.

OPTION 2 **`neighbouring_fact`**

MEANS a fact about an adjacent concept, disorder or event, where the families differ rather than the members

WHY HERE Having fixed on the two transitions that give the longest lines, n=2 for Lyman and n=3 for Balmer, a student reaches for the n^2 law that sits beside them in the same chapter - it belongs to the orbit radius, not to the emitted wavelength - and writes $2^2/3^2 = 4/9$. The wavelength needs $1/n_f^2 - 1/n_i^2$ inverted, which gives 5/27.

2015 Code A **Q137** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Electromagnetic Waves	Electromagnetic Spectrum	medium	2

1 The question

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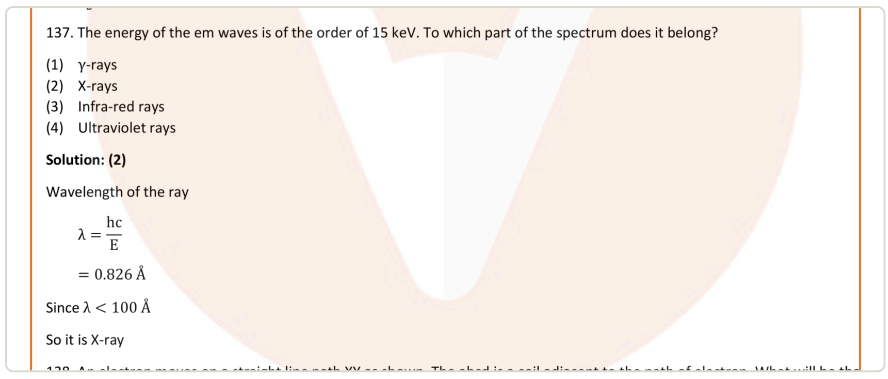

137. The energy of the em waves is of the order of 15 keV. To which part of the spectrum does it belong?

(1) γ-rays
(2) X-rays
(3) Infra-red rays
(4) Ultraviolet rays

Solution: (2)

Wavelength of the ray

$\lambda = \frac{hc}{E}$

$= 0.826$ Å

Since $\lambda < 100$ Å

So it is X-ray

READ FROM THE IMAGE BY B3-QUESTION-READER

The energy of the em waves is of the order of 15 keV. To which part of the spectrum does it belong?

(1) γ-rays
(2) X-rays ←printed answer
(3) Infra-red rays
(4) Ultraviolet rays

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **E = hc / lambda**

APPROACH Convert 15 keV to joules, invert E = hc/lambda for the wavelength, and match that wavelength against the bands of the electromagnetic spectrum.

ALSO USES **the EM spectrum bands are fixed by wavelength range**

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
2	2 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT Physics

CHAPTER Electromagnetic Waves HIGH CONFIDENCE

WHY THIS CHAPTER The answer turns on where a 15 keV photon (about 0.8 angstrom) sits in the electromagnetic spectrum, which is the energy/wavelength ordering of the spectrum taught in Electromagnetic Waves.

SUBTOPIC 1 Electromagnetic Spectrum

WHY THIS SUBTOPIC The correct answer depends on placing a 15 keV photon in the spectrum's energy ordering, which is exactly the electromagnetic-spectrum topic.

DIFFICULTY **medium** · 2 steps

STEPS convert 15 keV to a wavelength using E = hc/lambda (about 0.8 angstrom) → place that wavelength in the electromagnetic spectrum and name the band

WHY One energy-to-wavelength substitution followed by locating the result on the spectrum.

THE CATCH Gamma rays sit immediately beyond X-rays in energy, so the candidate must actually place 15 keV on the scale rather than merely know the ordering of the bands.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

plain_mcq

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

name_or_identify

MEANS a name, term, colour, organism, scientist

E3 Content shape

what is being tested, plus the actual concept

example_list

MEANS a taught list or table of named items each tied to a trait, role, effect or partner; membership or the pairing IS the fact, and nothing about how it works rebuilds it

> *Electromagnetic spectrum: the photon-energy decade each band occupies - a few eV for ultraviolet, tens of keV for X-rays, an MeV and above for*
>
> *γ*
>
> *-rays*

E4 Thinking

three separate questions, not one

recall

MEANS one stored fact decides it

single_step

MEANS one step, however much knowledge it needs

elimination

MEANS name the one pair or fact and the options it

E5 Trap

one option, one mistake, or nothing

OPTION 1 **other_one_fact**

MEANS a fact about a different member of the same family — another hormone, another organelle, another syndrome

WHY HERE γ-rays are the band a student files "high energy" under, and 15 keV reads as high energy. The γ decade starts around an MeV, some seventy times higher, while tens of keV is the working range of an X-ray tube.

OPTION 4 **unit_or_conversion**

MEANS a missing or wrong conversion yields it

WHY HERE Reading 15 keV as 15 eV - dropping the k - lands squarely in the ultraviolet, where a few electron-volts is the right scale. The k is a factor of a thousand and carries the photon into the X-ray decade.

OPTION 3 **no_derivation_route**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE Infra-red photons are of order 10^{-2} eV, six decades below the printed 15 keV. The two prefix misreadings this stem allows - k read as nothing, giving 15 eV, and k read as M, giving 15 MeV - land on options 4 and 1, and no single misreading of one prefix spans six decades.

2015 Code A **Q138** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Electromagnetic Induction	Direction of Induced Current	hard	4

1 The question

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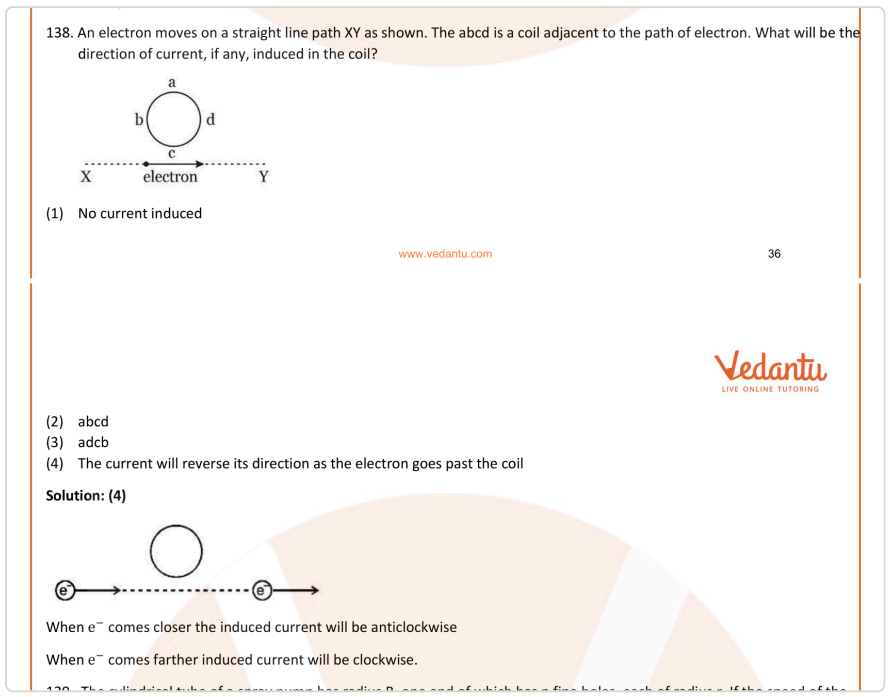

138. An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil?

(1) No current induced

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(2) abcd
(3) adcb
(4) The current will reverse its direction as the electron goes past the coil

Solution: (4)

When e^- comes closer the induced current will be anticlockwise

When e^- comes farther induced current will be clockwise.

READ FROM THE IMAGE BY B3-QUESTION-READER

An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil?

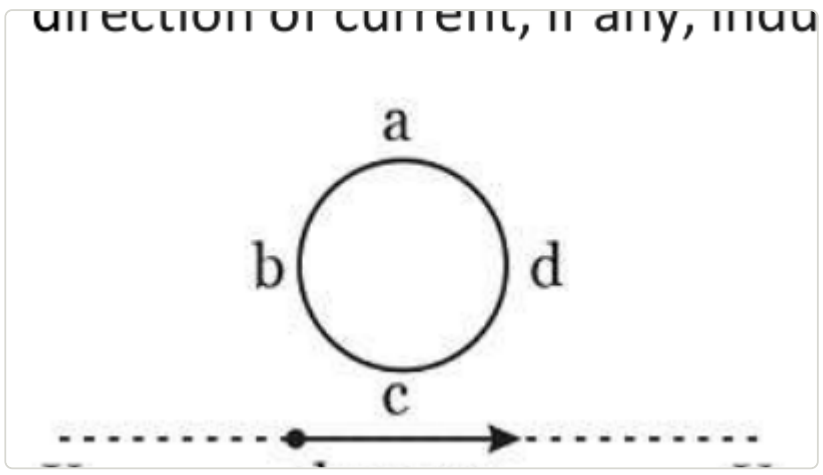


figure reported by the agent · `papers_v2/figures_agent/2015_A/q138.png`

(1) No current induced

(2) abcd

(3) adcb

(4) **The current will reverse its direction as the electron goes past the coil** ←printed answer

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT

Lenz's law: the induced current opposes the change in flux, e = -dPhi/dt

APPROACH

The moving electron's flux through the coil grows as it approaches and falls as it recedes, so apply Lenz's law to each half of the pass separately to fix the sense of the induced current.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
4	4 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Physics
CHAPTER	Electromagnetic Induction HIGH CONFIDENCE
WHY THIS CHAPTER	The answer turns on Lenz's law fixing the sense of the induced current from whether the electron's flux through the coil is growing or decaying.
SUBTOPIC 1	Direction of Induced Current
SUBTOPIC 2	Lenz's Law
WHY THIS SUBTOPIC	The answer is the sense of the induced current, fixed by Lenz's law from the flux of the moving electron growing then decaying as it passes the coil.
DIFFICULTY	**hard** · 4 steps
STEPS	translate the moving electron into an equivalent conventional current, which points opposite to the electron's velocity → find the direction of the magnetic field that current sets up through the plane of the coil → apply Lenz's law over the approach, while the flux through the coil is growing → apply Lenz's law again after the electron has passed, while the flux is falling, and compare the two induced directions

WHY The work is two separate Lenz's-law determinations, one on each side of the coil, after converting the electron's motion into an equivalent current direction.

THE CATCH The carrier is negative, so the equivalent current runs against the motion, and three options assume one steady direction - which is exactly what you get by analysing only the approach and stopping there.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

plain_mcq
MEANS a question and four answers, no lettered items, no lists, no scheme

figure_to_read
MEANS the question cannot be answered without reading a figure

E2 Format

what the ANSWER must contain

pick_direction
MEANS a direction or sense — the direction of a magnetic field, a force, or an angular acceleration (eastward, clockwise, into the page)

E3 Content shape

what is being tested, plus the actual concept

formula_or_law
MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Lenz's law when a flux rises and then falls: the induced current opposes the change, so its sense flips as the moving charge passes the coil*

E4 Thinking

three separate questions, not one

application
MEANS a stated rule, criterion or convention applied to this case. The rule must be known first — application is not knowledge-free

multi_step_dependent
MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

symmetry

E5 Trap

one option, one mistake, or nothing

OPTION 2 **intermediate_value**
MEANS a correct intermediate result is printed as an option

WHY HERE Working out the sense while the electron is still approaching - the into-page flux through the coil growing, so the induced current runs abcd - is the first half of the answer, and a student who has it in hand reads it off. The electron then moves away, the same flux falls, and the current runs the other way.

OPTION 3 **sign_reversed_magnitude_right**
MEANS the magnitude and the method are right and one sign, right-hand rule or inequality is inverted

WHY HERE Treating the moving electron as a conventional current in its own direction of travel - forgetting the charge is negative - puts the flux out of the page while it approaches, and Lenz's law then returns adcb. The geometry and the method are right; only the sign of the charge is dropped.

2015 Code A **Q139** PHYSICS

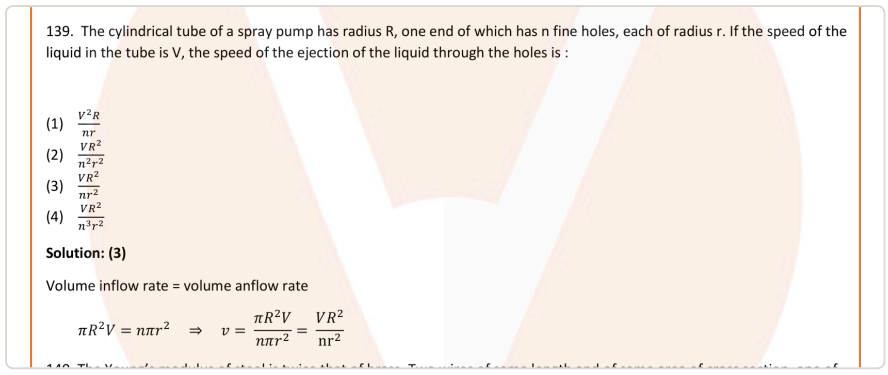
concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Mechanical Properties of Fluids	Equation of Continuity and Bernoulli's Equation	medium	3

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

PRINTED IN THE BOOKLET



139. The cylindrical tube of a spray pump has radius R, one end of which has n fine holes, each of radius r. If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes is :

(1) $\frac{V^2R}{nr}$
(2) $\frac{VR^2}{n^2r^2}$
(3) $\frac{VR^2}{nr^2}$
(4) $\frac{VR^2}{n^3r^2}$

Solution: (3)

Volume inflow rate = volume anflow rate

$$\pi R^2 V = n\pi r^2 \quad \Rightarrow \quad v = \frac{\pi R^2 V}{n\pi r^2} = \frac{VR^2}{nr^2}$$

READ FROM THE IMAGE BY B3-QUESTION-READER

The cylindrical tube of a spray pump has radius R, one end of which has n fine holes, each of radius r. If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes is :

(1) $\frac{V^2R}{nr}$

(2) $\frac{VR^2}{n^2r^2}$

(3) $\mathbf{\frac{VR^2}{nr^2}}$ ←printed answer

(4) $\frac{VR^2}{n^3r^2}$

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **$A_1 v_1 = A_2 v_2$ (equation of continuity)**

APPROACH Equate the volume flow rate in the tube, pi R^2 V, to the combined rate through the n holes, n pi r^2 v, and solve for the ejection speed v.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Physics
CHAPTER	Mechanical Properties of Fluids HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the equation of continuity for incompressible flow, equating the tube's volume flow rate to that of the n holes.
SUBTOPIC 1	Equation of Continuity and Bernoulli's Equation
WHY THIS SUBTOPIC	The answer depends only on the equation of continuity, pi R ^ 2 V = n pi r ^ 2 v.
DIFFICULTY	**medium** · 3 steps
STEPS	apply the equation of continuity: the volume flow in the tube equals the total flow out of the holes → write the two areas, pi R ^ 2 for the tube and n times pi r ^ 2 for the n holes together → solve for the ejection speed
WHY	The work is one conservation law written down, the hole area multiplied by n, and a rearrangement.
THE CATCH	Option 2 is the exact result of squaring n along with r when writing the combined hole area.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`derive_expression`

MEANS a symbolic expression

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Equation of continuity with* n *outlets:* $\pi R^2 V = n\pi r^2 v$ *, so the ejection speed is* $VR^2/(nr^2)$

E4 Thinking

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`single_step`

MEANS one step, however much knowledge it needs

`dimensional_check`

`elimination`

MEANS name the one pair or fact and the options it

E5 Trap

one option, one mistake, or nothing

OPTION 2 **`wrong_power_of_dependence`**

MEANS linear where it should be squared, squared where it should be $\sqrt{ed}$, or a root dropped

WHY HERE Treating n holes as one hole of n times the radius puts $(nr)^2$ in the denominator in place of $n\ r^2$. It is the hole area that multiplies by n, not the radius, so n belongs at the first power and not the second.

OPTION 4 **`no_derivation_route`**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE n enters continuity exactly once, through the total outlet area $n\pi\ r^2$, so the only slips available are putting it inside the square - which gives option 2's n^2 - or leaving it out, which gives n^0 and is not printed. Nothing reaches n^3; it only extends the printed set into a ladder of n^1, n^2, n^3.

OPTION 1 **`no_derivation_route`**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE The nearest route to the shape R/(nr) is using the tube's circumference in place of its cross-section, $2\pi\ R\ V = n\ 2\pi\ r\ v$, and that gives VR/(nr) - one power of V short of what is printed. $V^2R/(nr)$ is not a speed at all.

2015 Code A **Q141** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Current Electricity	Potentiometer	medium	3

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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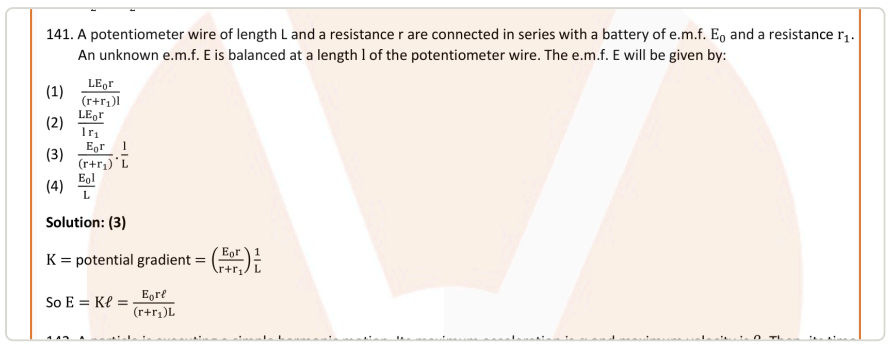


141. A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. E is balanced at a length l of the potentiometer wire. The e.m.f. E will be given by:

(1) $\frac{LE_0r}{(r+r_1)l}$

(2) $\frac{LE_0r}{l\,r_1}$

(3) $\frac{E_0r}{(r+r_1)}\cdot\frac{l}{L}$

(4) $\frac{E_0l}{L}$

Solution: (3)

K = potential gradient $= \left(\frac{E_0r}{r+r_1}\right)\frac{1}{L}$

So $E = K\ell = \frac{E_0r\ell}{(r+r_1)L}$

READ FROM THE IMAGE BY B3-QUESTION-READER

A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. E is balanced at a length l of the potentiometer wire. The e.m.f. E will be given by:

(1) $\frac{LE_0r}{(r+r_1)l}$

(2) $\frac{LE_0r}{l\,r_1}$

(3) $\frac{E_0r}{(r+r_1)}\cdot\frac{l}{L}$ ←printed answer

(4) $\frac{E_0l}{L}$

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **potentiometer balance: E = k l, k = potential gradient of the wire**

APPROACH Ohm's law fixes the drop across the wire as $E_0\, r/(r + r_1)$; divide by L for the gradient k, then E = k l gives the unknown e.m.f. at balance length l.

ALSO USES **V = IR in the series loop**

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Physics
CHAPTER	Current Electricity HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the potentiometer's potential gradient, i.e. the drop E0*r/(r+r1) distributed along the wire, which is the series-circuit/Ohm's-law result taught in Current Electricity.
SUBTOPIC 1	Potentiometer
WHY THIS SUBTOPIC	At balance the unknown emf equals the wire's potential gradient times the balancing length, and that gradient is the driving current E0/(r+r1) times r, spread over length L, giving E = E0*r*l/((r+r1)*L).
DIFFICULTY	**medium** · 3 steps
STEPS	find the current in the driver loop, I = E0/(r + r1) → get the potential drop across the whole potentiometer wire, I*r, and divide by L for the potential gradient → set the unknown emf equal to the gradient times the balancing length l
WHY	A three-step derivation: loop current, potential gradient along the wire, then the balance condition at length l.
THE CATCH	The option E0*l/L is exactly what a candidate gets by forgetting that r1 and r divide E0, and another option inverts l and L, so both common slips land on printed options.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`derive_expression`

MEANS a symbolic expression

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Potentiometer balance: the driving cell sends* $E_0/(r+r_1)$ *through the wire, setting a gradient* $E_0 r/[(r+r_1)L]$*, and the unknown emf is that gradient times the balancing length*

E4 Thinking

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`elimination`

MEANS name the one pair or fact and the options it

E5 Trap

one option, one mistake, or nothing

OPTION 1 **`inverted_ratio`**

MEANS the reciprocal of the right answer is printed

WHY HERE Setting the balance up as a proportion - the whole wire is to the balanced length as the whole drop is to E - puts L/l where the gradient needs l/L. Everything else in option 1 matches the key, so only the length fraction is upside down, and it makes E grow as the balance point moves back towards the start of the wire.

OPTION 4 **`omitted_contribution`**

MEANS one force, field contribution, circuit element or term is left out of an otherwise correct sum

WHY HERE The potentiometer is first met as "E is the fraction of the wire balanced times the driving emf", E_0 l/L, which is the result for a wire that has the whole cell across it. Here r_1 and r divide that emf, so only $E_0 r/(r+r_1)$ ever reaches the wire.

2015 Code A **Q142** PHYSICS

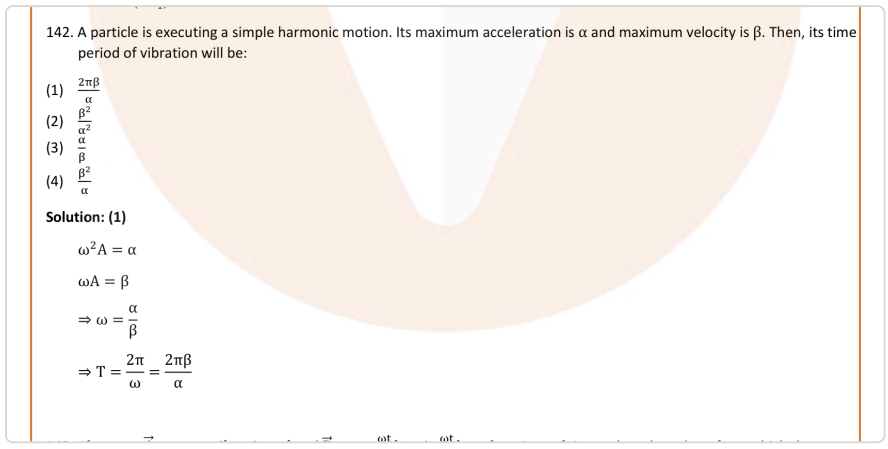
concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Oscillations	Calculate Time Period of SHM	medium	1

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

PRINTED IN THE BOOKLET


142. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be:

(1) $\frac{2\pi\beta}{\alpha}$

(2) $\frac{\beta^2}{\alpha^2}$

(3) $\frac{\alpha}{\beta}$

(4) $\frac{\beta^2}{\alpha}$

Solution: (1)

$\omega^2 A = \alpha$

$\omega A = \beta$

$\Rightarrow \omega = \frac{\alpha}{\beta}$

$\Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi\beta}{\alpha}$

READ FROM THE IMAGE BY B3-QUESTION-READER

A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be:

(1) $\frac{2\pi\beta}{\alpha}$ ←printed answer

(2) $\frac{\beta^2}{\alpha^2}$

(3) $\frac{\alpha}{\beta}$

(4) $\frac{\beta^2}{\alpha}$

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **SHM: a_max = w² A, v_max = w A**

APPROACH Divide the maximum acceleration by the maximum velocity to get w, then T = 2 pi / w gives the time period in terms of the two given maxima.

ALSO USES **T = 2 pi / w**

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT Physics

CHAPTER Oscillations HIGH CONFIDENCE

WHY THIS CHAPTER The answer depends on the SHM relations a_max = omega^2 A and v_max = omega A, whose ratio fixes omega and hence the time period.

SUBTOPIC 1 Calculate Time Period of SHM

WHY THIS SUBTOPIC The answer depends on extracting omega from a_max/v_max = omega and converting it to the time period T = 2 pi / omega = 2 pi beta / alpha.

DIFFICULTY **medium** · 3 steps

STEPS write the maximum acceleration as omega squared times amplitude and set it equal to alpha → write the maximum speed as omega times amplitude and set it equal to beta → divide the two to eliminate the amplitude, giving omega, and substitute into T = 2 pi / omega

WHY Two standard SHM maxima written down, then one elimination of the unknown amplitude and a substitution into the period formula.

THE CATCH One option is alpha/beta, which is the angular frequency itself - the exact result of stopping one step short of the period.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`derive_expression`

MEANS a symbolic expression

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *SHM: maximum acceleration is* $\omega^2 A$ *and maximum speed is* ωA*, so their ratio gives* ω *and hence the period*

E4 Thinking

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`dimensional_check`

E5 Trap

one option, one mistake, or nothing

OPTION 3 **`intermediate_value`**

MEANS a correct intermediate result is printed as an option

WHY HERE A student who divides α by β, gets ω and reads that as the answer has stopped at the first step. The period is still 2π/ω.

OPTION 4 **`wrong_quantity`**

MEANS the option answers for a neighbouring PHYSICAL quantity: diameter for radius, moment of inertia for radius of gyration

WHY HERE Combining the same two relations the other way round gives β^2/α, which is the amplitude. A student who eliminates without tracking which symbol survives answers for the wrong quantity in the same problem.

OPTION 2 **`wrong_power_of_dependence`**

MEANS linear where it should be squared, squared where it should be $\sqrt{ed}$, or a root dropped

WHY HERE Squaring the ratio is the reflex when one relation already carries ω^2, as if the square had to be cancelled by another square. $(\beta/\alpha)^2$ varies as a time squared, so it cannot be a period.

2015 Code A **Q143** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Kinematics	Dot Product	medium	4

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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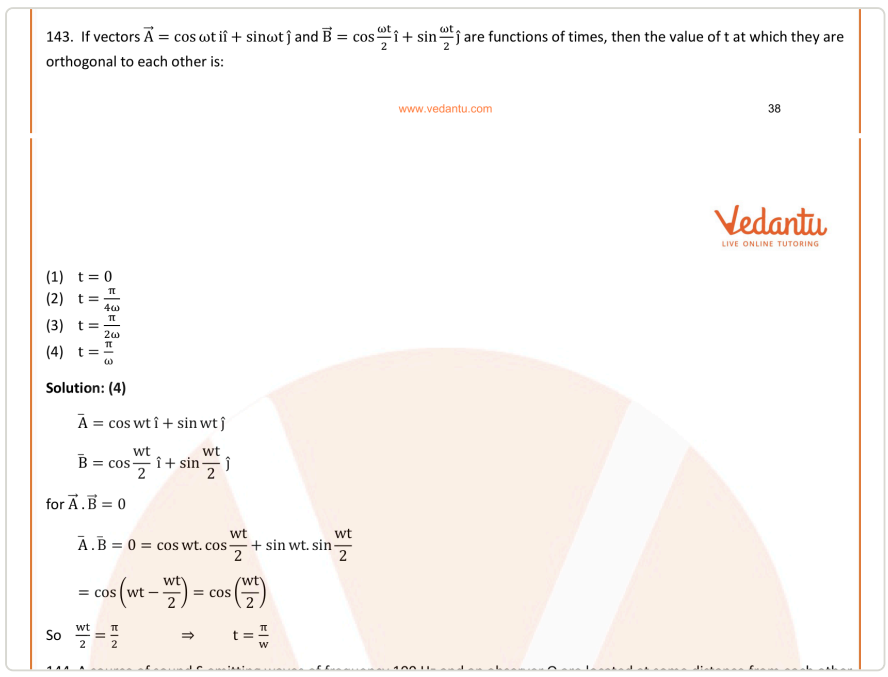
143. If vectors $\vec{A} = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}$ are functions of times, then the value of t at which they are orthogonal to each other is:

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(1) $t = 0$
(2) $t = \frac{\pi}{4\omega}$
(3) $t = \frac{\pi}{2\omega}$
(4) $t = \frac{\pi}{\omega}$

Solution: (4)

$\vec{A} = \cos wt\,\hat{i} + \sin wt\,\hat{j}$

$\vec{B} = \cos\frac{wt}{2}\,\hat{i} + \sin\frac{wt}{2}\,\hat{j}$

for $\vec{A}.\vec{B} = 0$

$\vec{A}.\vec{B} = 0 = \cos wt.\cos\frac{wt}{2} + \sin wt.\sin\frac{wt}{2}$

$= \cos\left(wt - \frac{wt}{2}\right) = \cos\left(\frac{wt}{2}\right)$

So $\frac{wt}{2} = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{w}$

READ FROM THE IMAGE BY B3-QUESTION-READER

If vectors $\vec{A} = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\,\hat{i} + \sin\frac{\omega t}{2}\,\hat{j}$ are functions of times, then the value of t at which they are orthogonal to each other is:

(1) $t = 0$

(2) $t = \frac{\pi}{4\omega}$

(3) $t = \frac{\pi}{2\omega}$

(4) $t = \frac{\pi}{\omega}$ ←printed answer

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **A . B = 0 for orthogonal vectors**

APPROACH Dot the two vectors; the cos-cos plus sin-sin terms collapse by the cosine difference identity, and setting that to zero solves for t.

ALSO USES **cos A cos B + sin A sin B = cos(A - B)**

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
4	4 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Physics
CHAPTER	Kinematics HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the scalar (dot) product being zero for perpendicular vectors, giving cos(wt/2) = 0 and t = pi/w.
SUBTOPIC 1	Dot Product
WHY THIS SUBTOPIC	The correct answer depends on setting A.B = cos(wt/2) = 0 for orthogonality, giving t = pi/w.
DIFFICULTY	**medium** · 3 steps
STEPS	set the dot product of the two vectors to zero as the condition for orthogonality → expand the dot product and collapse it with the cosine difference identity to cos(omega t / 2) → solve cos(omega t / 2) = 0 for the smallest positive t
WHY	The work is one orthogonality condition, one trigonometric identity, and solving the resulting equation.
THE CATCH	Option 3 is exactly what comes out of solving omega t = pi/2 instead of omega t / 2 = pi/2, so dropping the half-angle lands on an offered value.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`calculate_value`

MEANS a number the student computes from the stem's numbers

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

Two vectors are perpendicular when their dot product is zero;

$\cos\omega\, t\cos\frac{\omega\, t}{2} + \sin\omega\, t\sin\frac{\omega\, t}{2}$

collapses to

$\cos\frac{\omega\, t}{2}$

E4 Thinking

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`elimination`

MEANS name the one pair or fact and the options it

E5 Trap

one option, one mistake, or nothing

OPTION 3 **`calculation_slip`**

MEANS a specific wrong operation yields a printed option

WHY HERE A student who reduces the dot product to cos(ω t/2) and then solves it the way cosω t = 0 is always solved sets ω t = π/2. That gives t = π/2ω, half the correct time.

OPTION 1 **`no_derivation_route`**

MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE Zero is the first value to try in a question asking for a time, and at t = 0 the two vectors look symmetric. They are in fact both along î, so the dot product is 1 and no route makes it vanish there.

2015 Code A **Q144** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Waves	Doppler Effect	medium	3

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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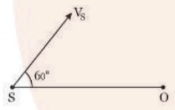
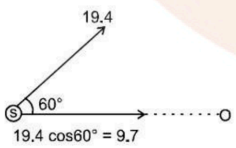
144. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms^{-1} at an angle of 60^o with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms^{-1}), is:



(1) 97 Hz
(2) 100 Hz
(3) 103 Hz
(4) 106 Hz

Solution: (3)



19.4 cos60° = 9.7

$t^1 = f_0\left(\frac{v - v}{v - v_s}\right)$

$f^1 = 100\left(\frac{v - 0}{v - (+9.7)}\right)$

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$f^1 = 100\frac{v}{v\left(1 - \frac{9.7}{v}\right)}$

$f^1 = 100\left(1 + \frac{3.7}{330}\right) = 103Hz$

READ FROM THE IMAGE BY B3-QUESTION-READER

A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms^{-1} at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms^{-1}), is:

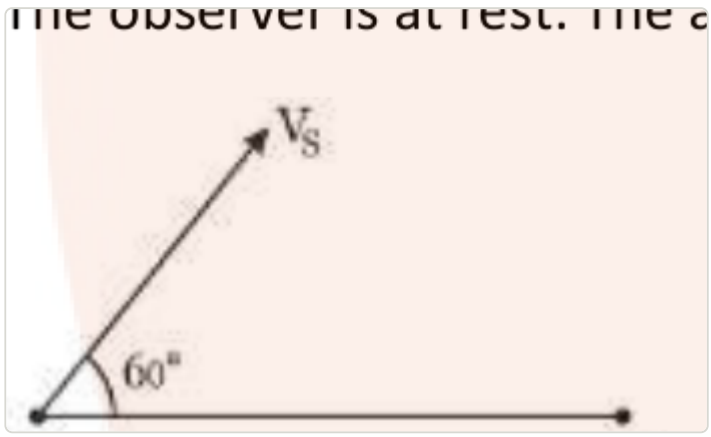

figure reported by the agent · `papers_v2/figures_agent/2015_A/q144.png`

(1) 97 Hz

(2) 100 Hz

(3) 103 Hz ←printed answer

(4) 106 Hz

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT

f' = f v / (v - v_s cos(theta)) (Doppler, source moving at an angle)

APPROACH

Only the component of the source velocity along the source-observer line shifts the pitch, so resolve v_s through 60 degrees and put it into the moving-source Doppler relation to get the apparent frequency.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Physics
CHAPTER	Waves HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the Doppler formula for a moving source, with only the component of the source velocity along the source-observer line counting.
SUBTOPIC 1	Doppler Effect
WHY THIS SUBTOPIC	Only the component of the source's velocity along the source-observer line, 19.4*cos60 = 9.7 m/s, enters the Doppler formula, giving f' = 100*330/(330-9.7) = 103 Hz.
DIFFICULTY	**medium** · 3 steps
STEPS	see from the figure that only the component of the source velocity along the source-observer line matters, and that it points towards the observer → resolve it: 19.4 cos 60 = 9.7 m/s → substitute into the moving-source Doppler formula f' = f v/(v - vs cos theta) and evaluate
WHY	A resolution of the velocity along the line of sight, then one substitution into the Doppler relation with the approaching sign.

THE CATCH 97 Hz is on offer as the exact result of taking the source as receding (wrong sign in the denominator), and 100 Hz as the result of treating the 60 degree motion as purely transverse.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`
MEANS a question and four answers, no lettered items, no lists, no scheme

`figure_to_read`
MEANS the question cannot be answered without reading a figure

E2 Format

what the ANSWER must contain

`calculate_value`
MEANS a number the student computes from the stem's numbers

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`
MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Doppler effect with motion at an angle: only the component of the source's velocity along the source-observer line shifts the frequency*

E4 Thinking

three separate questions, not one

`calculation`
MEANS arriving at a number or expression by computing

`multi_step_dependent`
MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`elimination`
MEANS name the one pair or fact and the options it

`estimation`

E5 Trap

one option, one mistake, or nothing

OPTION 1
`sign_reversed_magnitude_right`
MEANS the magnitude and the method are right and one sign, right-hand rule or inequality is inverted

WHY HERE A student who writes the Doppler formula with the plus sign in the denominator has recalled the receding case. The 3 Hz shift is the right size and lands on the wrong side: 330/(330+9.7) gives 97 Hz.

OPTION 2 **`copied_from_stem`**
MEANS the option repeats a phrase from the stem without answering the question

WHY HERE "Motion across the line gives no shift" is a real result for purely transverse motion, and a student who applies it here hands back the 100 Hz printed in the stem. The 60° motion still has a component pointing at the observer.

OPTION 4 **`condition_ignored`**
MEANS the option is what you get by dropping a stated condition — "battery disconnected", "in acidic medium", the specified temperature range

WHY HERE The formula is learned for a source moving straight at the observer, where the whole speed goes in, so the printed 19.4 m/s is used as it stands. That gives 106 Hz; resolving it to 9.7 m/s gives 103 Hz.

2015 Code A **Q146** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Magnetic Effect of Electric Current	Torque on Current Loop	hard	3

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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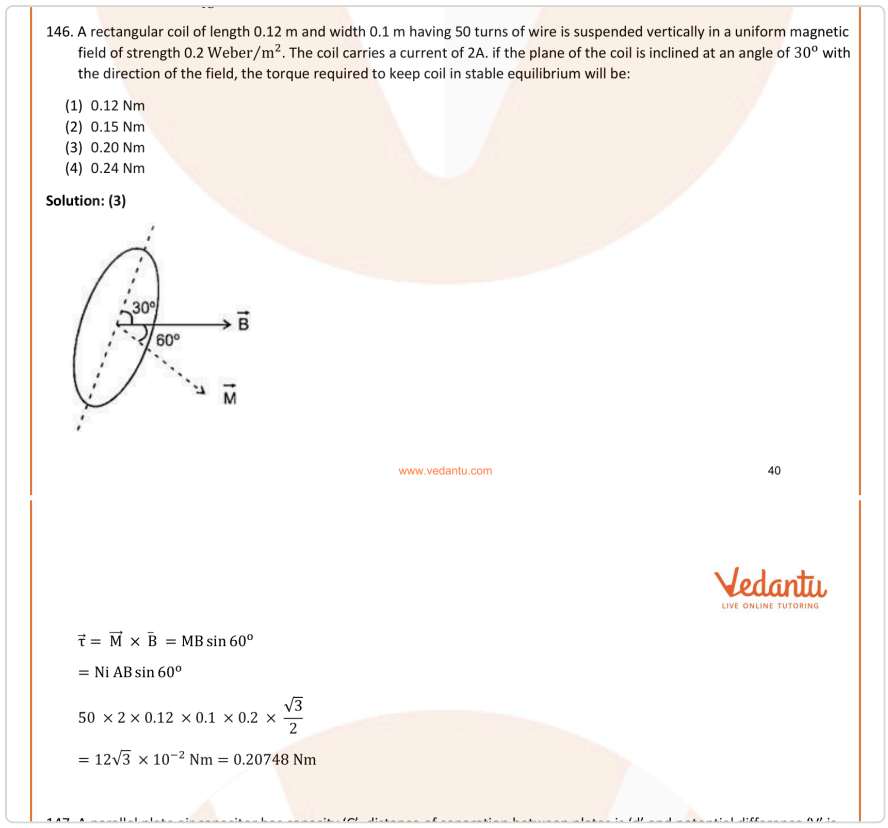
146. A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m^2. The coil carries a current of 2A. if the plane of the coil is inclined at an angle of 30^o with the direction of the field, the torque required to keep coil in stable equilibrium will be:

(1) 0.12 Nm
(2) 0.15 Nm
(3) 0.20 Nm
(4) 0.24 Nm

Solution: (3)

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$\vec{\tau} = \vec{M} \times \vec{B} = MB \sin 60^o$

$= Ni\,AB \sin 60^o$

$50 \times 2 \times 0.12 \times 0.1 \times 0.2 \times \frac{\sqrt{3}}{2}$

$= 12\sqrt{3} \times 10^{-2}\ \text{Nm} = 0.20748\ \text{Nm}$

READ FROM THE IMAGE BY B3-QUESTION-READER

A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m². The coil carries a current of 2A. if the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep coil in stable equilibrium will be:

(1) 0.12 Nm

(2) 0.15 Nm

(3) 0.20 Nm ←printed answer

(4) 0.24 Nm

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **tau = N I A B sin(theta), theta between the coil normal M and B**

APPROACH A plane tilted 30 degrees to B puts the normal at 60 degrees to it, so with A = length x width the relation gives the torque needed to hold the coil.

ALSO USES **M = N I A**

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT Physics

CHAPTER Magnetic Effect of Electric Current HIGH CONFIDENCE

WHY THIS CHAPTER The answer depends on the torque on a current-carrying loop in a uniform magnetic field, tau = NIAB sin(angle between the normal and B).

SUBTOPIC 1 Torque on Current Loop

WHY THIS SUBTOPIC The correct answer depends on tau = NIAB sin(theta) for a current loop, with theta measured from the normal so the plane's 30 degree tilt gives a cos30 factor.

DIFFICULTY **hard** · 4 steps

STEPS compute the coil area from the given length and width → form the magnetic moment of the coil as N times I times A → resolve the geometry - the plane is inclined at 30 degrees to the field, so the normal makes 60 degrees with it, and the torque is N I A B sin(60) → multiply out to the numerical torque

WHY An area, a magnetic moment, a geometric conversion from the plane's angle to the normal's angle, and a final product.

THE CATCH One option is N I A B with the angle dropped altogether and another is N I A B sin(30), which is exactly what comes out of treating the stated 30 degrees as the angle from the normal; the plane-versus-normal distinction decides the whole question.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`calculate_value`

MEANS a number the student computes from the stem's numbers

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Torque on a current loop,*
>
> $\tau = NIAB\sin\theta$
>
> *with*
>
> θ
>
> *measured from the normal: a plane at*
>
> 30°
>
> *to the field puts the normal at*
>
> 60°

E4 Thinking

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`elimination`

MEANS name the one pair or fact and the options it

E5 Trap

one option, one mistake, or nothing

OPTION 4 **`condition_ignored`**

MEANS the option is what you get by dropping a stated condition — "battery disconnected", "in acidic medium", the specified temperature range

WHY HERE NIAB is the form of the torque formula that gets memorised, with the angle factor treated as a detail. 0.24 Nm is the right answer only when the field lies in the plane of the coil.

OPTION 1 **`calculation_slip`**

MEANS a specific wrong operation yields a printed option

WHY HERE The formula is learned as NIABsinθ, so the printed 30° goes straight into the sine. That angle is measured from the plane, which puts the normal at 60°: $0.24\cos 30^\circ = 0.20$, not $0.24\sin 30^\circ = 0.12$.

2015 Code A **Q147** PHYSICS

concept and approach recorded

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Physics	Capacitance	The Parallel Plate Capacitor	medium	3

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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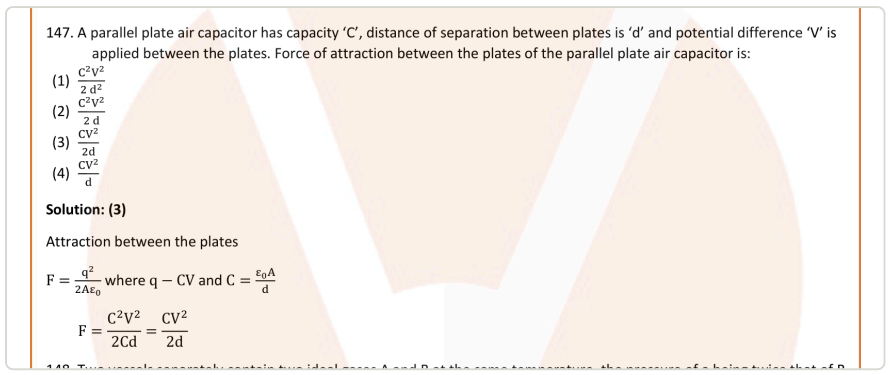

147. A parallel plate air capacitor has capacity 'C', distance of separation between plates is 'd' and potential difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is:

(1) $\frac{C^2V^2}{2d^2}$
(2) $\frac{C^2V^2}{2d}$
(3) $\frac{CV^2}{2d}$
(4) $\frac{CV^2}{d}$

Solution: (3)

Attraction between the plates

$F = \frac{q^2}{2A\varepsilon_0}$ where $q - CV$ and $C = \frac{\varepsilon_0 A}{d}$

$F = \frac{C^2V^2}{2Cd} = \frac{CV^2}{2d}$

READ FROM THE IMAGE BY B3-QUESTION-READER

A parallel plate air capacitor has capacity 'C', distance of separation between plates is 'd' and potential difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is:

(1) $\frac{C^2V^2}{2d^2}$

(2) $\frac{C^2V^2}{2d}$

(3) $\frac{CV^2}{2d}$ ←printed answer

(4) $\frac{CV^2}{d}$

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **F = q² / (2 epsilono A), with q = CV and C = epsilono A / d**

APPROACH Write the plate attraction in terms of the charge, then substitute q = CV and eliminate epsilono A using C = epsilono A / d to express the force through C, V and d.

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
3	3 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Physics
CHAPTER	Capacitance HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the force between parallel plates, obtained from the energy stored in a parallel plate capacitor (or from each plate sitting in half the inter-plate field), expressed through C, V and d.
SUBTOPIC 1	The Parallel Plate Capacitor
SUBTOPIC 2	Energy Stored in a Capacitor
WHY THIS SUBTOPIC	The answer depends on the attractive force between parallel plates, F = CV^2/(2d), which follows from the stored energy U = CV^2/2 varying with the plate separation.
DIFFICULTY	**medium** · 3 steps
STEPS	write the charge on each plate as Q = CV → take the field acting on one plate as that due to the other plate alone, V/(2d), not the full field V/d between the plates → multiply charge by that field to get the attractive force in terms of C, V and d
WHY	The work is charge from capacitance, the field one plate feels from the other, and their product.
THE CATCH	Option 4 is exactly the result of using the full inter-plate field V/d instead of the half-field one plate experiences, so the factor of two the question is built around has a ready-made wrong option.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 **Question type**

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 **Format**

what the ANSWER must contain

`derive_expression`

MEANS a symbolic expression

E3 **Content shape**

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

> *Force between capacitor plates: each plate sits in half the field, so* $F = \frac{1}{2}CV^2/d$ *, the stored energy divided by the separation*

E4 **Thinking**

three separate questions, not one

`calculation`

MEANS arriving at a number or expression by computing

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`dimensional_check`

E5 **Trap**

one option, one mistake, or nothing

OPTION 4 **`calculation_slip`**

MEANS a specific wrong operation yields a printed option

WHY HERE Force as charge times field is the natural route, and the field between the plates is V/d, so a student multiplies CV by it. Each plate sits in only the other plate's half of that field, so this comes out twice the answer.

OPTION 1 **`wrong_power_of_dependence`**

MEANS linear where it should be squared, squared where it should be $\sqrt{ed}$, or a root dropped

WHY HERE A student who reaches for the Coulomb shape - charge squared over distance squared - writes $(CV)^2/2d^2$. The force goes as C once and as 1/d once, and a charge squared over an area is not a force at all without $1/\varepsilon_0$.

OPTION 2 **`wrong_power_of_dependence`**

MEANS linear where it should be squared, squared where it should be $\sqrt{ed}$, or a root dropped

WHY HERE Writing the energy as $\frac{1}{2}QV$ and then putting Q = CV in without dropping the C already there counts the capacitance twice. One factor of C too many.

2016 Code X **Q55** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Principles of Inheritance and Variation	Pleiotropy, Multiple Alleles & Polygenic Inheritance	medium	1

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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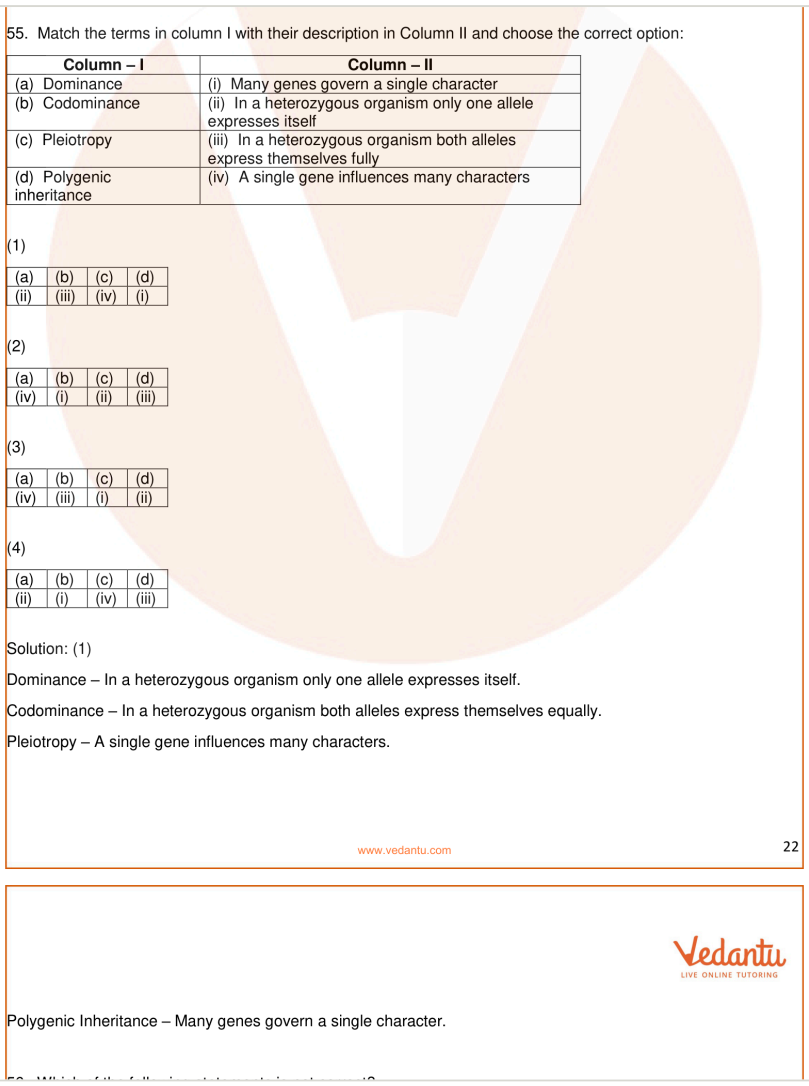

55. Match the terms in column I with their description in Column II and choose the correct option:

Column – I	Column – II
(a) Dominance	(i) Many genes govern a single character
(b) Codominance	(ii) In a heterozygous organism only one allele expresses itself
(c) Pleiotropy	(iii) In a heterozygous organism both alleles express themselves fully
(d) Polygenic inheritance	(iv) A single gene influences many characters

(1)

(a)	(b)	(c)	(d)
(ii)	(iii)	(iv)	(i)

(2)

(a)	(b)	(c)	(d)
(iv)	(i)	(ii)	(iii)

(3)

(a)	(b)	(c)	(d)
(iv)	(iii)	(i)	(ii)

(4)

(a)	(b)	(c)	(d)
(ii)	(i)	(iv)	(iii)

Solution: (1)

Dominance – In a heterozygous organism only one allele expresses itself.

Codominance – In a heterozygous organism both alleles express themselves equally.

Pleiotropy – A single gene influences many characters.

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Polygenic Inheritance – Many genes govern a single character.

READ FROM THE IMAGE BY B3-QUESTION-READER

Match the terms in column I with their description in Column II and choose the correct option:

(a)	Dominance	(i)	Many genes govern a single character
(b)	Codominance	(ii)	In a heterozygous organism only one allele expresses itself
(c)	Pleiotropy	(iii)	In a heterozygous organism both alleles express themselves fully
(d)	Polygenic inheritance	(iv)	A single gene influences many characters

(1) (a) (ii) (b) (iii) (c) (iv) (d) (i) ←printed answer

(2) (a) (iv) (b) (i) (c) (ii) (d) (iii)

(3) (a) (iv) (b) (iii) (c) (i) (d) (ii)

(4) (a) (ii) (b) (i) (c) (iv) (d) (iii)

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

how NCERT reaches this answer: **states answer**

NEW NCERT · BIO12L CH4 · PRINTED PAGE 006

genotype **t**. As a result of random fertilisation, the resultant zygotes can be of the genotypes **TT**, **Tt** or **tt.**

From the Punnett square it is easily seen that $1/4^{th}$ of the random fertilisations lead to **TT**, 1/2 lead to **Tt** and $1/4^{th}$ to **tt**. Though the F_1 have a genotype of **Tt**, but the phenotypic character seen is 'tall'. At F_2, $3/4^{th}$ of the plants are tall, where some of them are **TT** while others are **Tt**. Externally it is not possible to distinguish between the plants with the genotypes **TT** and **Tt**. Hence, within the genopytic pair **Tt** only one character '**T**' tall is expressed. Hence the character **T** or 'tall' is said to dominate over the other allele **t** or 'dwarf' character. It is thus due to this dominance of one character over the other that all the F_1 are tall (though the genotype is **Tt**) and in the F_2 $3/4^{th}$ of the plants are tall (though genotypically 1/2 are **Tt** and only $1/4^{th}$ are **TT**). This leads to a phenotypic ratio of $3/4^{th}$ tall : (1/4 **TT** + 1/2 **Tt**) and $1/4^{th}$ **tt,** i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1.

The 1/4 : 1/2 : 1/4 ratio of **TT**: **Tt**: **tt** is mathematically condensable to the form of the binomial expression $(ax + by)^2$, that has the gametes bearing genes **T** or **t** in equal frequency of ½. The expression is expanded as given below :

$$(1/2\mathbf{T} + 1/2\,\mathbf{t})^2 = (1/2\mathbf{T} + 1/2\mathbf{t}) \times (1/2\mathbf{T} + 1/2\mathbf{t}) = 1/4\,\mathbf{TT} + 1/2\mathbf{Tt} + 1/4\,\mathbf{tt}$$

Mendel self-pollinated the F_2 plants and found that dwarf F_2 plants continued to generate dwarf plants in F_3 and F_4 generations. He concluded that the genotype of the dwarfs was homozygous – **tt**. What do you think he would have got had he self-pollinated a tall F_2 plant?

From the preceeding paragraphs it is clear that though the genotypic ratios can be calculated using mathematical probability, by simply looking at the phenotype of a dominant trait, it is not possible to know the genotypic composition. That is, for example, whether a tall plant from F_1 or F_2 has **TT** or **Tt** composition, cannot be predicted. Therefore, to determine the genotype of a tall plant at F_2, Mendel crossed the tall plant from F_2 with a dwarf plant. This he called a **test cross**. In a typical test cross an organism (pea plants here) showing a dominant phenotype (and whose genotype is to be determined) is crossed with the recessive parent instead of self-crossing. The progenies of such a cross can easily be analysed to predict the genotype of the test organism. Figure 4.5 shows the results of typical test cross where violet colour flower (V) is dominant over white colour flower (v).

Using Punnett square, try to find out the nature of offspring of a test cross.

What ratio did you get?

Using the genotypes of this cross, can you give a general definition for a test cross?

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BIO12L-CH04-P24 · 4.2 INHERITANCE OF ONE GENE

> From the Punnett square it is easily seen that 1/4th of the random fertilisations lead to TT, 1/2 lead to Tt and 1/4th to tt. Though the F1 have a genotype of Tt, but the phenotypic character seen is 'tall'. At F2, 3/4th of the plants are tall, where some of them are TT while others are Tt. Externally it is not possible to distinguish between the plants with the genotypes TT and Tt. Hence, within the genopytic pair Tt only one character

'T' tall is expressed. Hence the character T or 'tall' is said to dominate over the other allele t or 'dwarf' character. It is thus due to this dominance of one character over the other that all the F1 are tall (though the genotype is Tt) and in the F2 3/4th of the plants are tall (though genotypically 1/2 are Tt and only 1/4th are TT). This leads to a phenotypic ratio of 3/4th tall : (1/4 TT + 1/2 Tt) and 1/4th tt, i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1.

In the first case, the modified allele is equivalent to the unmodified allele, i.e., it will produce the same phenotype/trait, i.e., result in the transformation of substrate S. Such equivalent allele pairs are very common. But, if the allele produces a non-functional enzyme or no enzyme, the phenotype may be effected. The phenotype/trait will only be dependent on the functioning of the unmodified allele. The unmodified (functioning) allele, which represents the original phenotype is the dominant allele and the modified allele is generally the recessive allele. Hence, in the example above the recessive trait is seen due to non-functional enzyme or because no enzyme is produced.

4.2.2.2 Co-dominance

Till now we were discussing crosses where the F_1 resembled either of the two parents (dominance) or was in-between (incomplete dominance). But, in the case of co-dominance the F_1 generation resembles both parents. A good example is different types of red blood cells that determine ABO blood grouping in human beings. ABO blood groups are controlled by the gene ***I***. The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene. The gene **(*I*)** has three alleles I^A, I^B and ***i***. The alleles I^A and I^B produce a slightly different form of the sugar while allele ***i*** does not produce any sugar. Because humans are diploid organisms, each person possesses any two of the three ***I*** gene alleles. I^A and I^B are completely dominant over ***i***, in other words when I^A and ***i*** are present only I^A expresses (because ***i*** does not produce any sugar), and when I^B and ***i*** are present I^B expresses. But when I^A and I^B are present together they both express their own types of sugars: this is because of co-dominance. Hence red blood cells have both A and B types of sugars. Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types (Table 4.2). *How many phenotypes are possible?*

Table 4.2: Table Showing the Genetic Basis of Blood Groups in Human Population

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood types of offspring
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	$I^A i$	A
I^B	I^A	$I^A I^B$	AB
I^B	I^B	$I^B I^B$	B
I^B	i	$I^B i$	B
i	i	$i\,i$	O

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Reprint 2026-27

BIO12L-CH04-P39 · 4.2.2.2 Co-dominance

> 4.2.2.2 Co-dominance Till now we were discussing crosses where the F1 resembled either of the two parents (dominance) or was in-between (incomplete dominance). But, in the case of co-dominance the F1 generation resembles both parents. A good example is different types of red blood cells that determine ABO blood grouping in human beings. ABO blood groups are controlled by the gene I. The plasma membrane of the red blood

cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene. The gene (I) has three alleles I^A, I^B and i. The alleles I^A and I^B produce a slightly different form of the sugar while allele i does not produce any sugar. Because humans are diploid organisms, each person possesses any two of the three I gene alleles. I^A and I^B are completely dominant over i, in other words when I^A and i are present only I^A expresses (because i does not produce any sugar), and when I^B and i are present I^B expresses. But when I^A and I^B are present together they both express their own types of sugars: this is because of co-dominance. Hence red blood cells have both A and B types of sugars. Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types (Table 4.2). How many phenotypes are possible?

4.4 Polygenic Inheritance

Mendel's studies mainly described those traits that have distinct alternate forms such as flower colour which are either purple or white. But if you look around you will find that there are many traits which are not so distinct in their occurrence and are spread across a gradient. For example, in humans we don't just have tall or short people as two distinct alternatives but a whole range of possible heights. Such traits are generally controlled by three or more genes and are thus called as polygenic traits. Besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment. Human skin colour is another classic example for this. In a polygenic trait the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive. To understand this better let us assume that three genes A, B, C control skin colour in human with the dominant forms A, B and C responsible for dark skin colour and the recessive forms a, b and c for light skin colour. The genotype with all the dominant alleles (AABBCC) will have the darkest skin colour and that with all the recessive alleles (aabbcc) will have the lightest skin colour. As expected the genotype with three dominant alleles and three recessive alleles will have an intermediate skin colour. In this manner the number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.

4.5 Pleiotropy

We have so far seen the effect of a gene on a single phenotype or trait. There are however instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes. An example of this is the disease phenylketonuria, which occurs in humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

4.6 Sex Determination

The mechanism of sex determination has always been a puzzle before the geneticists. The initial clue about the genetic/chromosomal mechanism of sex determination can be traced back to some of the experiments carried out in insects. In fact, the cytological observations made in a number of insects led to the development of the concept of genetic/chromosomal basis of sex-determination. Henking (1891) could trace a specific nuclear structure all through spermatogenesis in a few insects, and it was also observed by him that 50 per cent of the sperm received this structure after spermatogenesis, whereas the other 50 per cent sperm did not receive it. Henking gave a name to this structure as the **X body** but he could not explain its significance. Further investigations by other scientists led to the conclusion that the 'X body' of Henking was in fact a chromosome

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BIO12L-CH04-P67 · 4.5 PLEIOTROPY

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> humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

NEW NCERT · BIO12L CH4 · PRINTED PAGE 017

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BIO12L-CH04-P66 · 4.4 POLYGENIC INHERITANCE

> 4.4 POLYGENIC INHERITANCE Mendel's studies mainly described those traits that have distinct alternate forms such as flower colour which are either purple or white. But if you look around you will find that there are many traits which are not so distinct in their occurrence and are spread across a gradient. For example, in humans we don't just have tall or short people as two distinct alternatives but a whole range of possible heights.

Such traits are generally controlled by three or more genes and are thus called as polygenic traits. Besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment. Human skin colour is another classic example for this. In a polygenic trait the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive. To understand this better let us assume that three genes A, B, C control skin colour in human with the dominant forms A, B and C responsible for dark skin colour and the recessive forms a, b and c for light skin colour. The genotype with all the dominant alleles (AABBCC) will have the darkest skin colour and that with all the recessive alleles (aabbcc) will have the lightest skin colour. As expected the genotype with three dominant alleles and three recessive alleles will have an intermediate skin colour. In this manner the number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 3 FOUND

2015 Code A · Q90	A pleiotropic gene:	easy · answer 1
2015 Code A · Q107	A gene showing codominance has:	easy · answer 1
2020 Code F1 · Q48	Identify the wrong statement with reference to the gene ‘I’ that controls ABO blood groups.	unrated

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Principles of Inheritance and Variation HIGH CONFIDENCE
WHY THIS CHAPTER	The correct answer depends on the definitions of dominance, codominance, pleiotropy and polygenic inheritance as deviations from simple Mendelian expression.
SUBTOPIC 1	Pleiotropy, Multiple Alleles & Polygenic Inheritance
SUBTOPIC 2	Incomplete Dominance & Co-dominance
WHY THIS SUBTOPIC	The answer depends on keeping the direction of the gene-to-trait relation straight - pleiotropy is one gene affecting many characters while polygenic inheritance is many genes governing one character - and on co-dominance meaning both alleles express fully in the heterozygote, which is what separates option 1 from the three scrambled keys.
DIFFICULTY	**medium** · 3 steps
STEPS	recall that dominance means only one allele expresses itself in a heterozygote, fixing (a) → recall that codominance means both alleles express themselves fully in a heterozygote, fixing (b) → separate pleiotropy (one gene affecting many characters) from polygenic inheritance (many genes governing one character) and keep the single code consistent with all four pairs
WHY	The work is three definitional recalls read against the code options, with the last two needing to be told apart because they are near-mirror statements.
THE CATCH	Descriptions (i) and (iv) are deliberate mirror images - many genes to one character versus one gene to many characters - so reversing them produces a code that is also on offer.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`match_the_lists`

MEANS two labelled lists to pair, answer is a pairing code

E2 Format

what the ANSWER must contain

`pick_match_code`

MEANS a pairing code (A-II, B-IV, ...)

E3 Content shape

what is being tested, plus the actual concept

`definition_term`

MEANS what a term or quantity means, including holding two neighbouring terms apart; the strict reading of the definition decides it

> *Dominance, codominance, pleiotropy and polygenic inheritance: which definition belongs to which term, and which way the gene-to-character relation runs*

E4 **Thinking**

three separate questions, not one

`concept_discrimination`
MEANS choosing between two neighbouring concepts

`several_independent_checks`
MEANS each item verified separately, verdicts combined

`elimination`
MEANS name the one pair or fact and the options it

E5 **Trap**

one option, one mistake, or nothing

OPTION 2 **`no_derivation_route`**
MEANS no arithmetic or factual route reaches this option at all; it widens the spread or completes a grid

WHY HERE List II sorts itself into two families by its own opening words - (ii) and (iii) begin 'In a heterozygous organism', which can only describe an interaction between two alleles of one gene, while (i) and (iv) count genes against characters - and this option crosses both families, handing the two allele terms the gene-count descriptions and the two gene terms the allele descriptions. All four of its rows are refuted by the printed wording itself, so no partial reading of any single definition reaches this code.

OPTION 3 **`semantic_reversal`**
MEANS the relationship is reversed in words

WHY HERE Pleiotropy and polygenic inheritance are the same one-to-many relation read in opposite directions, so a student who stores 'pleiotropy has many in it' without storing which end is many takes (i) 'Many genes govern a single character' for it. Pleiotropy is one gene with many effects - the single phenylketonuria mutation behind both the mental retardation and the reduced pigmentation - and (c) (i) is printed in option 3 and nowhere else.

OPTION 4 **`wrong_pairing`**
MEANS one pair in a match code is swapped

WHY HERE Codominance and polygenic inheritance both carry the idea of more than one thing contributing, so a student holding only that much reads 'co-' as many genes acting together and polygeny's additive effect as both alleles expressing fully. Codominance is one gene whose two alleles both show in the heterozygote, as in the AB blood group, so this option is the key with rows (b) and (d) exchanged and rows (a) and (c) right.

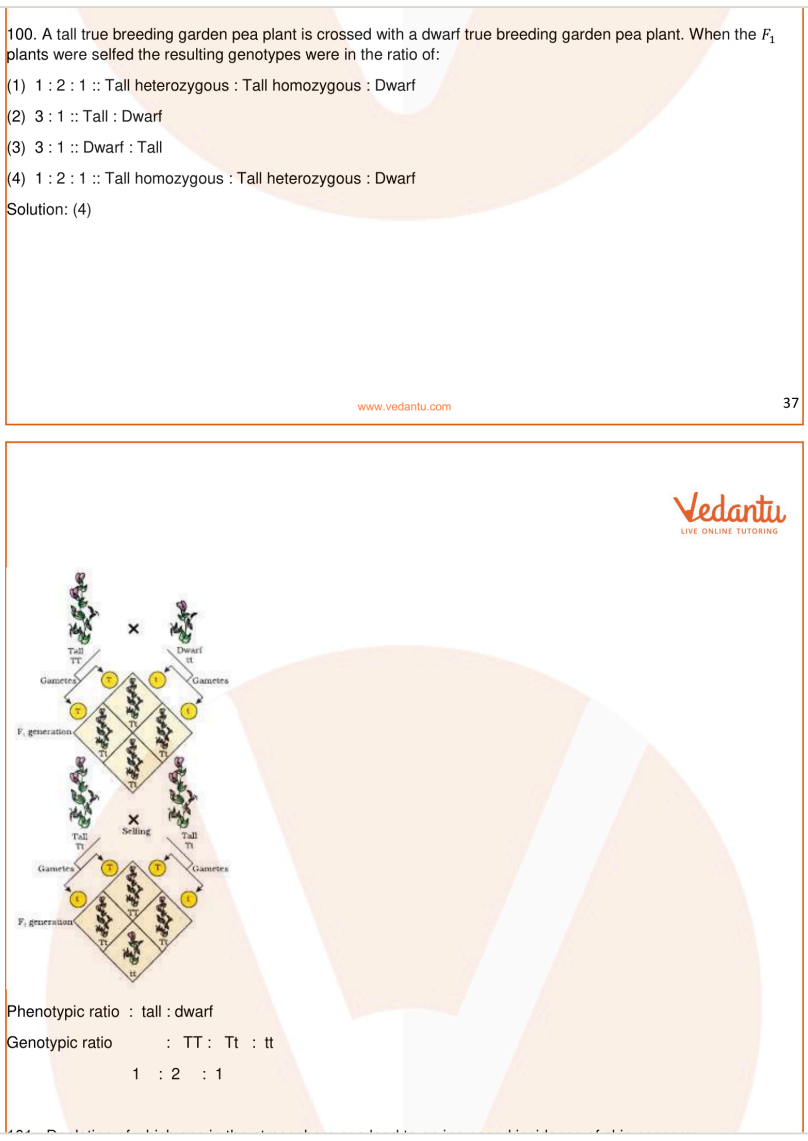
2016 Code X **Q100** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Principles of Inheritance and Variation	Monohybrid Cross	medium	4

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

PRINTED IN THE BOOKLET



READ FROM THE IMAGE BY B3-QUESTION-READER

A tall true breeding garden pea plant is crossed with a dwarf true breeding garden pea plant. When the F_1 plants were selfed the resulting genotypes were in the ratio of:

(1) 1 : 2 : 1 :: Tall heterozygous : Tall homozygous : Dwarf

(2) 3 : 1 :: Tall : Dwarf

(3) 3 : 1 :: Dwarf : Tall

(4) 1 : 2 : 1 :: Tall homozygous : Tall heterozygous : Dwarf ←printed answer

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT **1 : 2 : 1 genotypic ratio of a monohybrid F_2 (TT : Tt : tt)**

APPROACH TT x tt gives an all-heterozygous F_1, so selfing it gives the monohybrid F_2 square; genotypes are asked for, so the ratio is read against TT, Tt and tt in that order, not the phenotypic tall : dwarf split.

ALSO USES **law of segregation · 3 : 1 phenotypic ratio of a monohybrid F_2**

how NCERT reaches this answer: **states answer**

NEW NCERT · BIO12L CH4 · PRINTED PAGE 004

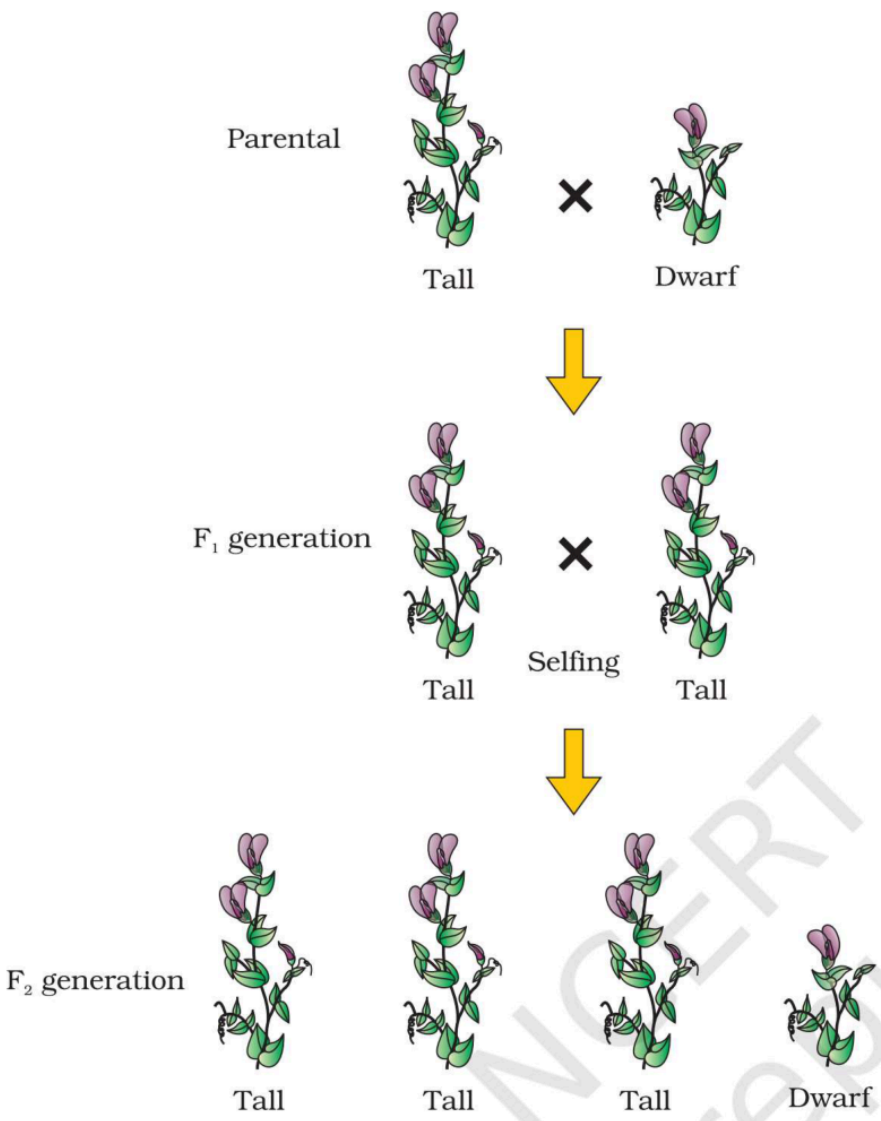


Figure 4.3 Diagrammatic representation of monohybrid cross

Based on these observations, Mendel proposed that something was being stably passed down, unchanged, from parent to offspring through the gametes, over successive generations. He called these things as 'factors'. Now we call them as **genes**. Genes, therefore, are the units of inheritance. They contain the information that is required to express a particular trait in an organism. Genes which code for a pair of contrasting traits are known as **alleles,** i.e., they are slightly different forms of the same gene.

If we use alphabetical symbols for each gene, then the capital letter is used for the trait expressed at the F_1 stage and the small alphabet for the other trait. For example, in case of the character of height, **T** is used for the Tall trait and **t** for the 'dwarf', and **T** and **t** are alleles of each other. Hence, in plants the pair of alleles for height would be **TT**, **Tt** or **tt**. Mendel also proposed that in a true breeding, tall or dwarf pea variety the allelic pair of genes for height are identical or **homozygous**, **TT** and **tt**, respectively. **TT** and **tt** are called the **genotype** of the plant while the descriptive terms **tall** and **dwarf** are the **phenotype**. What then would be the phenotype of a plant that had a genotype **Tt**?

As Mendel found the phenotype of the F_1 heterozygote **Tt** to be exactly like the **TT** parent in appearance, he proposed that in a pair of dissimilar factors, one dominates the other (as in the F_1) and hence is called the **dominant** factor while the other factor is **recessive** . In this case **T** (for tallness) is dominant over **t** (for dwarfness), that is recessive. He observed identical behaviour for all the other characters/trait-pairs that he studied.

It is convenient (and logical) to use the capital and lower case of an alphabetical symbol to remember this concept of dominance and recessiveness. (Do not use **T** for tall and **d** for dwarf because you will find it difficult to remember whether **T** and **d** are alleles of the same gene/character or not). Alleles can be similar as in the case of homozygotes **TT** and **tt** or can be dissimilar as in the case of the heterozygote **Tt**. Since

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BIO12L-CH04-P18 · 4.2 INHERITANCE OF ONE GENE

> If we use alphabetical symbols for each gene, then the capital letter is used for the trait expressed at the F1 stage and the small alphabet for the other trait. For example, in case of the character of height, T is used for the Tall trait and t for the 'dwarf', and T and t are alleles of each other. Hence, in plants the pair of alleles for height would be TT, Tt or tt. Mendel also proposed that in a true breeding, tall or dwarf pea variety the allelic pair of

> genes for height are identical or homozygous, TT and tt, respectively. TT and tt are called the genotype of the plant while the descriptive terms tall and dwarf are the phenotype. What then would be the phenotype of a plant that had a genotype Tt?

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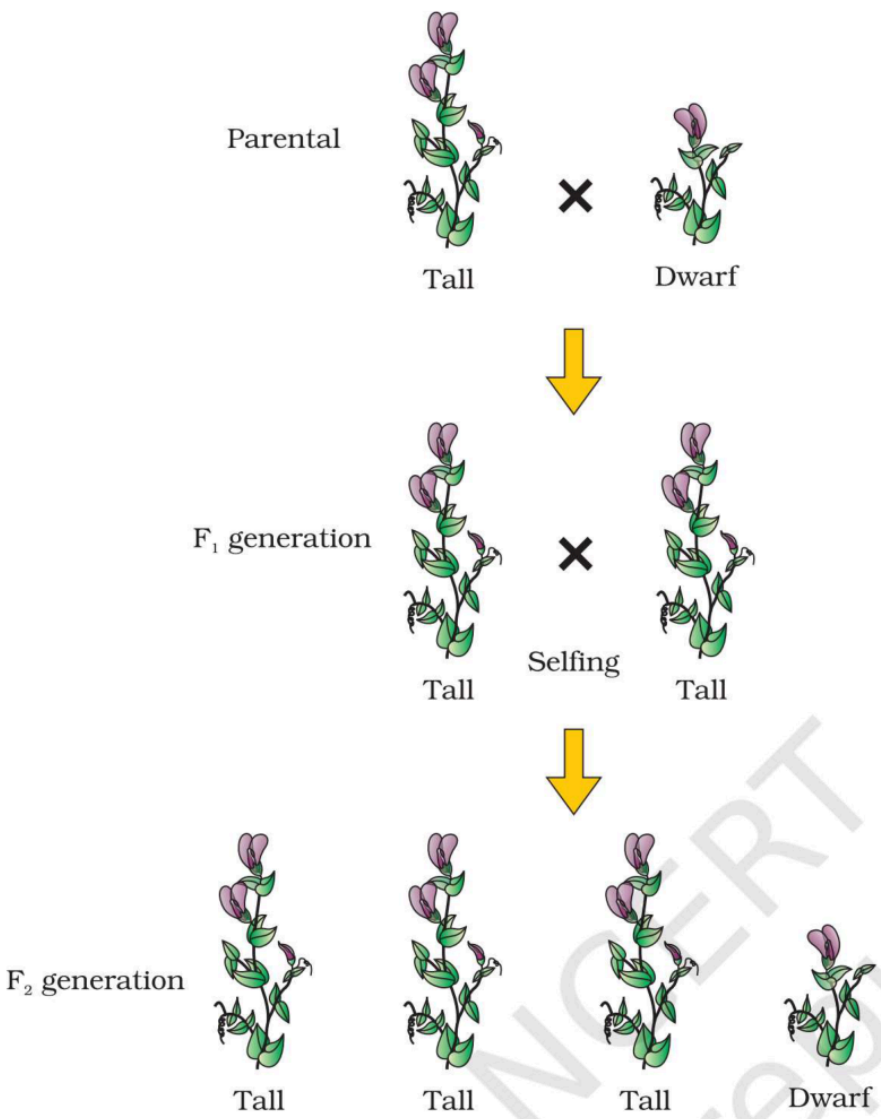

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If we use alphabetical symbols for each gene, then the capital letter is used for the trait expressed at the F_1 stage and the small alphabet for the other trait. For example, in case of the character of height, **T** is used for the Tall trait and **t** for the 'dwarf', and **T** and **t** are alleles of each other. Hence, in plants the pair of alleles for height would be **TT**, **Tt** or **tt**. Mendel also proposed that in a true breeding, tall or dwarf pea variety the allelic pair of genes for height are identical or **homozygous**, **TT** and **tt**, respectively. **TT** and **tt** are called the **genotype** of the plant while the descriptive terms **tall** and **dwarf** are the **phenotype**. What then would be the phenotype of a plant that had a genotype **Tt**?

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It is convenient (and logical) to use the capital and lower case of an alphabetical symbol to remember this concept of dominance and recessiveness. (Do not use **T** for tall and **d** for dwarf because you will find it difficult to remember whether **T** and **d** are alleles of the same gene/character or not). Alleles can be similar as in the case of homozygotes **TT** and **tt** or can be dissimilar as in the case of the heterozygote **Tt**. Since

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BIO12L-CH04-P20 · 4.2 INHERITANCE OF ONE GENE

> It is convenient (and logical) to use the capital and lower case of an alphabetical symbol to remember this concept of dominance and recessiveness. (Do not use T for tall and d for dwarf because you will find it difficult to remember whether T and d are alleles of the same gene/character or not).

> Alleles can be similar as in the case of homozygotes TT and tt or can be dissimilar as in the case of the heterozygote Tt. Since the Tt plant is heterozygous for genes controlling one character (height), it is a monohybrid and the cross between TT and tt is a monohybrid cross.

NEW NCERT · BIO12L CH4 · PRINTED PAGE 006

genotype **t**. As a result of random fertilisation, the resultant zygotes can be of the genotypes **TT**, **Tt** or **tt.**

From the Punnett square it is easily seen that $1/4^{th}$ of the random fertilisations lead to **TT**, 1/2 lead to **Tt** and $1/4^{th}$ to **tt**. Though the F_1 have a genotype of **Tt**, but the phenotypic character seen is 'tall'. At F_2, $3/4^{th}$ of the plants are tall, where some of them are **TT** while others are **Tt**. Externally it is not possible to distinguish between the plants with the genotypes **TT** and **Tt**. Hence, within the genopytic pair **Tt** only one character '**T**' tall is expressed. Hence the character **T** or 'tall' is said to dominate over the other allele **t** or 'dwarf' character. It is thus due to this dominance of one character over the other that all the F_1 are tall (though the genotype is **Tt**) and in the F_2 $3/4^{th}$ of the plants are tall (though genotypically 1/2 are **Tt** and only $1/4^{th}$ are **TT**). This leads to a phenotypic ratio of $3/4^{th}$ tall : (1/4 **TT** + 1/2 **Tt**) and $1/4^{th}$ **tt,** i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1.

The 1/4 : 1/2 : 1/4 ratio of **TT**: **Tt**: **tt** is mathematically condensable to the form of the binomial expression $(ax + by)^2$, that has the gametes bearing genes **T** or **t** in equal frequency of ½. The expression is expanded as given below :

$$(1/2\mathbf{T} + 1/2\,\mathbf{t})^2 = (1/2\mathbf{T} + 1/2\mathbf{t}) \times (1/2\mathbf{T} + 1/2\mathbf{t}) = 1/4\,\mathbf{TT} + 1/2\mathbf{Tt} + 1/4\,\mathbf{tt}$$

Mendel self-pollinated the F_2 plants and found that dwarf F_2 plants continued to generate dwarf plants in F_3 and F_4 generations. He concluded that the genotype of the dwarfs was homozygous – **tt**. What do you think he would have got had he self-pollinated a tall F_2 plant?

From the preceeding paragraphs it is clear that though the genotypic ratios can be calculated using mathematical probability, by simply looking at the phenotype of a dominant trait, it is not possible to know the genotypic composition. That is, for example, whether a tall plant from F_1 or F_2 has **TT** or **Tt** composition, cannot be predicted. Therefore, to determine the genotype of a tall plant at F_2, Mendel crossed the tall plant from F_2 with a dwarf plant. This he called a **test cross**. In a typical test cross an organism (pea plants here) showing a dominant phenotype (and whose genotype is to be determined) is crossed with the recessive parent instead of self-crossing. The progenies of such a cross can easily be analysed to predict the genotype of the test organism. Figure 4.5 shows the results of typical test cross where violet colour flower (V) is dominant over white colour flower (v).

Using Punnett square, try to find out the nature of offspring of a test cross.

What ratio did you get?

Using the genotypes of this cross, can you give a general definition for a test cross?

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BIO12L-CH04-P24 · 4.2 INHERITANCE OF ONE GENE

> From the Punnett square it is easily seen that 1/4th of the random fertilisations lead to TT, 1/2 lead to Tt and 1/4th to tt. Though the F1 have a genotype of Tt, but the phenotypic character seen is 'tall'. At F2, 3/4th of the plants are tall, where some of them are TT while others are Tt. Externally it is not possible to distinguish between the plants with the genotypes TT and Tt. Hence, within the genopytic pair Tt only one character

'T' tall is expressed. Hence the character T or 'tall' is said to dominate over the other allele t or 'dwarf' character. It is thus due to this dominance of one character over the other that all the F1 are tall (though the genotype is Tt) and in the F2 3/4th of the plants are tall (though genotypically 1/2 are Tt and only 1/4th are TT). This leads to a phenotypic ratio of 3/4th tall : (1/4 TT + 1/2 Tt) and 1/4th tt, i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 2 FOUND

2021 Code O1 · Q125	The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants can be understood from a diagram called	unrated · answer 1
2021 Code O1 · Q179	In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased?	unrated · answer 1

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
4	4 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Principles of Inheritance and Variation HIGH CONFIDENCE
WHY THIS CHAPTER	The correct answer depends on the F2 genotypic ratio of a monohybrid cross being 1 homozygous tall : 2 heterozygous tall : 1 dwarf, as distinct from the 3:1 phenotypic ratio.
SUBTOPIC 1	Monohybrid Cross
WHY THIS SUBTOPIC	The correct answer depends on the F2 genotypic ratio of a monohybrid cross being 1 homozygous tall : 2 heterozygous tall : 1 dwarf, not the 3:1 phenotypic ratio.
DIFFICULTY	**medium** · 2 steps
STEPS	work the monohybrid cross: true-breeding tall x true-breeding dwarf gives an all-heterozygous F1, and selfing that F1 gives genotypes in 1 : 2 : 1 → read the option labels in order and check which option attaches 1 to the homozygous tall, 2 to the heterozygous tall and 1 to the dwarf, since the stem asks for genotypes rather than phenotypes
WHY	One Punnett-square application followed by a careful read of which genotype each number in the printed ratio is assigned to.
THE CATCH	Option 1 carries the correct 1 : 2 : 1 ratio with homozygous and heterozygous swapped, and the 3 : 1 options are exactly the phenotypic answer a candidate gives who does not notice the stem says genotypes.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`pick_ratio`

MEANS a ratio, as a ratio

E3 Content shape

what is being tested, plus the actual concept

`formula_or_law`

MEANS a stated relation, law or rule applied, rearranged or chained — including a proportionality where constants cancel, a dimensional formula, and a limiting or boundary condition (critical angle, resonance, terminal velocity)

Monohybrid cross: the genotypic ratio

1 : 2 : 1

(TT : Tt : tt) against the phenotypic

3 : 1

E4 **Thinking**

three separate questions, not one

`application`

MEANS a stated rule, criterion or convention applied to this case. The rule must be known first — application is not knowledge-free

`multi_step_dependent`

MEANS step 2 needs step 1's result. Four independent match-checks are NOT four steps

`elimination`

MEANS name the one pair or fact and the options it

E5 **Trap**

one option, one mistake, or nothing

OPTION 1 **`wrong_pairing`**

MEANS one pair in a match code is swapped

WHY HERE A student who has the ratio right and reads the legend straight across, first class onto first number, puts the homozygote on the 2. The swapped pair is Tall homozygous against Tall heterozygous: Tt comes out of the square two ways and TT only one, so the heterozygote takes the 2 and the tall homozygote the 1.

OPTION 2 **`true_but_out_of_scope`**

MEANS the option is correct subject matter that answers a different question than the stem asked; it fails on relevance, not on truth

WHY HERE 3 : 1 is the number every monohybrid self is remembered by, so a student who answers from that number without registering the word "genotypes" takes it. It is true of this cross as the phenotypic ratio, and what the stem asked for was genotypes, where tall splits into TT and Tt.

OPTION 3 **`semantic_reversal`**

MEANS the relationship is reversed in words

WHY HERE The same remembered 3 : 1, with dominance put on the dwarf parent, gives 3 dwarf : 1 tall. Tallness is the dominant allele in this cross, so the class of three is the tall one - and the stem asked for genotypes in any case.

2016 Code X **Q126** BIOLOGY / BOTH verified against NCERT

SUBJECT	CHAPTER	SUBTOPIC	DIFFICULTY	ANSWER
Biology	Molecular Basis of Inheritance	DNA Fingerprinting	medium	1

1 The question

left: exactly as it is printed in the booklet. right: what we extracted from that image.

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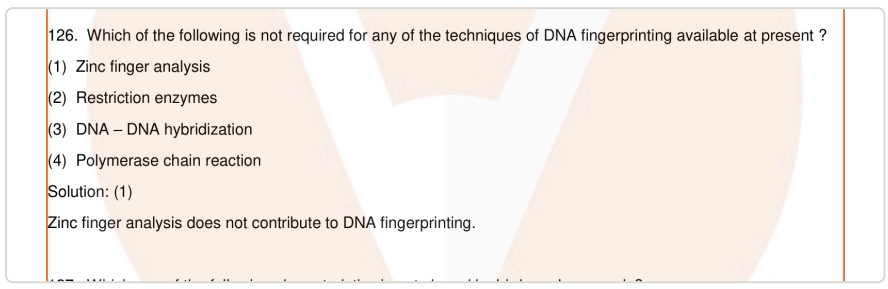
126. Which of the following is not required for any of the techniques of DNA fingerprinting available at present ?
(1) Zinc finger analysis
(2) Restriction enzymes
(3) DNA – DNA hybridization
(4) Polymerase chain reaction
Solution: (1)
Zinc finger analysis does not contribute to DNA fingerprinting.

READ FROM THE IMAGE BY B3-QUESTION-READER

Which of the following is not required for any of the techniques of DNA fingerprinting available at present ?

(1) Zinc finger analysis ←printed answer

(2) Restriction enzymes

(3) DNA – DNA hybridization

(4) Polymerase chain reaction

2 What the answer rests on

the NCERT page as printed, beside that page's extracted text with the quoted line marked. For Physics and Physical Chemistry, the concept and the approach instead — a numerical's answer is a value NCERT never prints.

CONCEPT

VNTR polymorphism read by restriction digestion, PCR and DNA-DNA hybridisation

APPROACH

Check each option against the steps a fingerprinting protocol actually runs - cutting, amplifying, probing - and single out the technique that belongs to protein-DNA binding work instead.

how NCERT reaches this answer: **concept applied**

the other members of population (through sexual reproduction). Allelic (again recall the definition of alleles from Chapter 4) sequence variation has traditionally been described as a DNA polymorphism if more than one variant (allele) at a locus occurs in human population with a frequency greater than 0.01. In simple terms, if an **inheritable mutation** is observed in a population at high frequency, it is referred to as **DNA polymorphism**. The probability of such variation to be observed in non-coding DNA sequence would be higher as mutations in these sequences may not have any immediate effect/impact in an individual's reproductive ability. These mutations keep on accumulating generation after generation, and form one of the basis of variability/polymorphism. There is a variety of different types of polymorphisms ranging from single nucleotide change to very large scale changes. For evolution and speciation, such polymorphisms play very important role, and you will study these in details at higher classes.

The technique of DNA Fingerprinting was initially developed by Alec Jeffreys. He used a satellite DNA as probe that shows very high degree of polymorphism. It was called as **Variable Number of Tandem Repeats** (VNTR). The technique, as used earlier, involved Southern blot hybridisation using radiolabelled VNTR as a probe. It included

(i) isolation of DNA,

(ii) digestion of DNA by restriction endonucleases,

(iii) separation of DNA fragments by electrophoresis,

(iv) transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon,

(v) hybridisation using labelled VNTR probe, and

(vi) detection of hybridised DNA fragments by autoradiography. A schematic representation of DNA fingerprinting is shown in Figure 5.16.

The VNTR belongs to a class of satellite DNA referred to as mini-satellite. A small DNA sequence is arranged tandemly in many copy numbers. The copy number varies from chromosome to chromosome in an individual. The numbers of repeat show very high degree of polymorphism. As a result the size of VNTR varies in size from 0.1 to 20 kb. Consequently, after hybridisation with VNTR probe, the autoradiogram gives many bands of differing sizes. These bands give a characteristic pattern for an individual DNA (Figure 5.16). It differs from individual to individual in a population except in the case of monozygotic (identical) twins. The sensitivity of the technique has been increased by use of polymerase chain reaction (PCR– you will study about it in Chapter 9). Consequently, DNA from a single cell is enough to perform DNA fingerprinting analysis. In addition to application in forensic science, it has much wider application, such as

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BIO12L-CH05-P139 · 5.10 DNA FINGERPRINTING

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you will study about it in Chapter 9). Consequently, DNA from a single cell is enough to perform DNA fingerprinting analysis. In addition to application in forensic science, it has much wider application, such as in determining population and genetic diversities. Currently, many different probes are used to generate DNA fingerprints.

HOW THE QUOTED LINES REACH THIS ANSWER

1. NCERT 5.10 names restriction endonucleases, Southern-blot/VNTR-probe hybridisation and PCR as parts of the DNA fingerprinting technique, so options 2, 3 and 4 are each required.
2. Zinc finger analysis is named nowhere in the chapter, so by elimination option 1 is the one not required.

THE SAME SUBTOPIC ELSEWHERE IN THESE PAPERS · 3 FOUND

2015 Code A · Q127	Satellite DNA is important because it :	medium · answer 3
2021 Code O1 · Q144	DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as:	unrated · answer 4
2022 Code Q1 · Q117	DNA polymorphism forms the basis of :	unrated · answer 3

3 The answer

the booklet's own printed answer, against an independent blind solve.

PRINTED IN THE BOOKLET	SOLVED INDEPENDENTLY	VERDICT
1	1 HIGH	agree

4 Where it belongs, and how hard it is

subject, branch, chapter and subtopic, each with the reason it was chosen.

SUBJECT	Biology
BRANCH	both
CHAPTER	Molecular Basis of Inheritance HIGH CONFIDENCE
WHY THIS CHAPTER	The answer depends on the actual steps of DNA fingerprinting (restriction digestion, hybridisation with VNTR probes, PCR); zinc finger analysis is not among them.
SUBTOPIC 1	DNA Fingerprinting
WHY THIS SUBTOPIC	The answer depends on what the technique actually uses - restriction digestion of DNA, Southern blotting with DNA-DNA hybridisation of VNTR probes, and PCR amplification of small samples - whereas zinc finger analysis studies a protein DNA-binding motif and has no role in any present fingerprinting protocol.
DIFFICULTY	**medium** · 2 steps
STEPS	recall the toolkit DNA fingerprinting actually uses: restriction digestion of the sample DNA, amplification by PCR, and hybridisation with labelled VNTR probes → screen the four options against that list and identify zinc finger analysis, a study of a DNA-binding protein motif, as the one that plays no part in any current fingerprinting technique
WHY	The negative stem means the recalled technique list has to be built first and then each option checked against it, rather than one fact being matched directly.
THE CATCH	Three of the four options are genuinely part of fingerprinting, so the odd one out is found only by recognising an unfamiliar term as belonging to protein chemistry rather than by elimination among familiar ones.

5 How it is categorised

the five fixed axes E1-E5. Each label shows what it MEANS and why this question has it.

E1 Question type

the shape on the page

`plain_mcq`

MEANS a question and four answers, no lettered items, no lists, no scheme

E2 Format

what the ANSWER must contain

`name_or_identify`

MEANS a name, term, colour, organism, scientist

E3 Content shape

what is being tested, plus the actual concept

`practical_procedure`

MEANS a named experiment, bench or field procedure, instrument or device: its setup, materials, what is in a named reagent, what an observation means, what each part is for

> *Steps of DNA fingerprinting: restriction digestion, electrophoresis, Southern blotting, hybridisation with a labelled VNTR probe and autoradiography, with PCR where the sample is small*

E4 Thinking

three separate questions, not one

recall

MEANS one stored fact decides it

several_independent_checks

MEANS each item verified separately, verdicts combined

elimination

MEANS name the one pair or fact and the options it

E5 Trap

one option, one mistake, or nothing

OPTION 2 **true_statement_trap**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE Present-day profiling amplifies short repeats by PCR and never cuts the sample, so a student working from current practice and the stem's "available at present" reads restriction enzymes as no longer needed. The stem asks what is not required for any of the available techniques, and the restriction-digest and Southern-blot method is still one of them.

OPTION 4 **true_statement_trap**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE The steps the chapter lists - isolation of DNA, digestion by restriction endonucleases, electrophoresis, blotting, hybridisation with a labelled probe and autoradiography - contain no PCR step, so a student reciting that protocol reads PCR as an extra and picks it. The same chapter requires PCR wherever the sample is too small to work with directly, which is most forensic material.

OPTION 3 **true_statement_trap**

MEANS the option is true but fails the task's polarity (NOT/EXCEPT)

WHY HERE DNA-DNA hybridization is also the name of the taxonomic method for measuring how closely two species are related, so a student with the phrase filed under phylogeny reads it as belonging to another subject altogether. It is the fingerprinting step itself: the labelled VNTR probe is hybridised to the blotted DNA, and without that step there are no bands to read.